PARTIAL DIFFERENTIAL EQUATIONS COURSE

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1. Introduction

Partial differential equations are used to model all physical situations. There can be no satisfactory general theory that applies to most PDEs. Changing an equation slightly can lead to completely different phenomenas. For example the 1D transport equation, the heat equation and the wave equation

$$u_t = u_x, \quad u_t = u_{xx}, \quad u_{tt} = u_{xx},$$

are obtained from one-another by replacing a first derivative by a second derivative, while the equation $u_t = -u_{xx}$ cannot be solved.

Instead, one needs to focus on classes of equations that have common features and try to develop a theory in each particular case, i.e PDEs from fluid mechanics, elasticity, gradient flows, dispersive equations etc.

It turns out that first order equations can be solved by ODEs by the method of characteristics (see Problem 1).

Elliptic equations.

The most basic second order equation which is coordinate independent is the Laplace equation

$$\triangle u := \sum u_{ii} = 0.$$

A constant coefficient second order equation $a^{ij} u_{ij} = 0$ can be reduced to Laplace's equation after an affine deformation if (and only if) the coefficient matrix $A = (a^{ij}) > 0$ is positive definite. Second order equations that are "governed by $\triangle$" are called elliptic equations

$$a^{ij}(x) u_{ij} + \text{lower order terms} = 0, \qquad A(x) = (a^{ij}(x)) > 0.$$

More generally, a second order equation

$$G(D^2 u, Du, u, x) = 0 \quad \text{is called } \textit{elliptic},$$

if $G(M, p, z, x)$ is increasing in the $M \in \mathcal{S}^{n\times n}$ variable.

Here $\mathcal{S}^{n\times n}$ is the space of symmetric matrices, and $M > N$ means that $M - N > 0$ is positive definite.

An evolution equation of the type

$$u_t = G(D^2 u, Du, u, x) \quad \text{is called } \textit{parabolic},$$

if G is elliptic. Most results for elliptic equations can be extended to parabolic equations.

In these notes we develop the basic theory for second order elliptic equations. One of the main goals is to understand the techniques involved in solving one of the most important second order PDEs which is the minimal surface equation. Several other nonlinear PDEs can be tackled by the same general principles.

Minimal Surface Equation.

We want to find a C^2 function $u : \overline{\Omega} \to \mathbb{R}$ so that $u = \varphi$ on $\partial\Omega$ with φ given, which minimizes the surface area of its graph

$$\mathcal{A}(u) = \int_\Omega \sqrt{1 + |\nabla u|^2} \, dx.$$

The Euler-Lagrange equation reads

$$div \frac{\nabla u}{\sqrt{1 + |\nabla u|^2}} = 0 \quad \text{in } \Omega,$$

or (using the summation convention)

$$\left(\delta_i^j - \frac{u_i u_j}{1 + |\nabla u|^2} \right) u_{ij} = 0,$$

or equivalently

$$\triangle u - \frac{u_i u_j}{1 + |\nabla u|^2} u_{ij} = 0.$$

In a more general form the equations above can be written as

$$a^{ij}(x, \nabla u) \, u_{ij} = 0, \qquad div(A(x, \nabla u)) = 0,$$

and they include more general minimiziation problems of the type

$$\text{minimize} \qquad \int_\Omega F(\nabla u) \, dx, \quad \text{with} \quad u = \varphi \quad \text{on} \quad \partial\Omega,$$

with $F : \mathbb{R}^n \to \mathbb{R}$ convex.

1.1. **Strategy for existence.**

A) Method of continuity:

If $u \in C^2$ satisfies

$$div \, (\nabla F(\nabla u)) = 0,$$

then it u is a minimizer of the energy, so it suffices to solve this equation in the class of C^2 functions (for a given boundary data).

Step 1. If $u \in C^2(\overline{\Omega})$ is a solution of the equation, then we show that we can also solve it as we perturb the boundary data to $u + \varepsilon\psi$. For this we need to develop the linear theory: We write the solution as $u + \varepsilon v$ and solve (the linearized equation for v)

$$div(D^2 F(\nabla u) \nabla v) = 0.$$

It turns out that the correct spaces (from a functional analytic point of view) to solve these linear equations are the Hölder spaces (Schauder theory).

Step 2. Apriori estimates: Show that $\|u\|_{C^{2,\alpha}} \leq C(\varphi, \Omega)$.

Steps 1 and 2 show that we can perturb the boundary data from a known solution (in this case say $u = 0$) to the one with data φ.

While Step 1 is standard and works in very general settings, Step 2 is the most crucial and it depends on the specific equation that we study.

B) Gradient flow: Solve the parabolic equation

$$u_t = div(DF(\nabla u)),$$

and let $t \to \infty$. In order to solve this, one needs to develop the theory of parabolic equations together with apriori estimates as well.

1.2. **Outline.** The plan is to develop first the linear theory (Schauder and Calderon-Zygmund estimates) for equations of the type

$$a^{ij}(x)\ u_{ij} = 0,$$

with coefficients $a^{ij}(x)$ continuous in x, by initially investigating the constant coefficient case of the Laplace equation $\triangle u = 0$.

Then we introduce Sobolev spaces and discuss the L^2 theory and the De Giorgi-Nash-Moser estimates for linear equations in divergence form

$$\partial_i(a^{ij}(x)\, u_j) = 0.$$

After this we can use apriori estimates and the method of continuity to solve non-linear problems like the minimal surface equation.

At the end we introduce viscosity solutions, the ABP estimate, and fully nonlinear elliptic equations.

1.3. **Problems.**

1. Assume that $u : \Omega \to \mathbb{R}$, $u \in C^2$ solves the 1st order PDE

$$H(Du, u, x) = 0,$$

with $H(p, z, x)$ a C^2 function. Fix a point x_0 and set $z_0 = u(x_0)$, $p_0 = Du(x_0)$.

a) Show that the derivative of Du at x_0 in the direction $D_pH(p_0, z_0, x_0)$ depends only on H and its derivatives at (p_0, z_0, x_0). The same is true for u and x.

b) Deduce that (Du, u, x) is determined uniquely along the "characteristic" ODE

$$p' = -D_xH - pD_zH, \qquad z' = p \cdot D_pH, \quad x' = D_pH,$$

with initial data (p_0, z_0, x_0).

2. Let $u(x,t) \in C^1(\mathbb{R} \times [0,T))$ solve the Burger's equation

$$u_t + uu_x = 0,$$

with initial boundary condition

$$u(x, 0) = \varphi(x).$$

Show the solution exists for all times ($T = \infty$) if $\varphi' \geq 0$, and exists only for a finite time ($T < \infty$) if φ' takes negative values.

3. Let $u \in C^1(\overline{B_1})$ solve

$$a(x) \cdot \nabla u + u = 0 \quad \text{in} \quad B_1.$$

Show that if $a(x) \cdot x > 0$ on ∂B_1, then $u \equiv 0$.

2. The Laplacian $\triangle u$

(Chapter 2 in Gilbarg-Trudinger [GT].)

For $u \in C^2$ we denote

$$\triangle u = u_{11} + ... + u_{nn} = div(\nabla u).$$

2.1. Coordinate independence. Assume $u \in C^2(\Omega) \cap C^1(\overline{\Omega})$ and $\partial\Omega \in C^1$.

1)

$$\int_\Omega \triangle u \, dx = \int_\Omega div(\nabla u)\, dx = \int_{\partial\Omega} u_\nu \; d\sigma.$$

2) ω_n - volume of the unit ball, $n\omega_n$-surface area of the unit sphere:

$$\frac{r}{n} \fint_{B_r} \triangle u \, dx = \fint_{\partial B_r} u_\nu \, d\sigma = \frac{d}{dr}\left(\fint_{\partial B_r} u \, d\sigma \right),$$

$$\fint_{B_r} \triangle u \, dx = \frac{n}{r}\frac{d}{dr}\left(\fint_{\partial B_r} u \, d\sigma \right).$$

3) If l is the linear approximation of u at 0 then

$$\triangle u(0) = \lim_{r \to 0^+} \frac{2n}{r^2} \fint_{\partial B_r} (u - l)\, d\sigma$$

$$= \lim_{r \to 0^+} \frac{c_n}{r^2} \fint_{B_r} (u - l)\, d\sigma,$$

with $c_n = 2(n+2)$.

2.2. Mean value property. Functions that satisfy $u \in C^2$ and $\triangle u = 0$ in Ω are called harmonic functions.

Examples of harmonic functions:

$$ax + by + c, \quad x^2 - y^2, \quad Re\, f(z), \quad r^\alpha \cos(\alpha\theta), \quad e^y \sin x.$$

In view of the property 2) above, the following are equivalent:

$$\triangle u = 0 \quad \Longleftrightarrow \quad u(x_0) = \fint_{\partial B_r(x_0)} u \, d\sigma \quad \Longleftrightarrow \quad u(x_0) = \fint_{B_r(x_0)} u \, dx,$$

(for any ball $B_r(x_0) \subset \Omega$).

The last property is called *mean value property* and the the one in the middle *spherical mean value property.*

For one of the implications we observe that if u has the mean value property then $u = u * \rho_\varepsilon \in C^2$ where ρ_ε is a symmetric mollifier.

2.3. The fundamental solution. We look for a radial solution $u(x) = g(r)$, with $r := |x|$ and obtain

$$\triangle u = g'' + \frac{n-1}{r} g' = 0.$$

We try $g(r) = r^\alpha$ and obtain the particular solution

$$\Gamma(x) := \begin{cases} c_n |x|^{2-n} & \text{if } n \geq 3, \text{ or} \\ \frac{-1}{2\pi} \log |x| & \text{if } n = 2. \end{cases}$$

The constant $c_n := \frac{1}{n(n-2)\omega_n}$ is such that (formally)

$$\int_{B_1} \triangle\Gamma \, dx = -1.$$

The last computation can be made more precise by defining the approximations

$$\Gamma_\rho(x) := \begin{cases} \Gamma, & \text{if } |x| \geq \rho \\ \\ P & \text{if } |x| < \rho, \end{cases}$$

with P the unique quadratic polynomial such that $\Gamma_\rho \in C^1$. Then

$$\triangle\Gamma_\rho = -\frac{1}{|B_\rho|}\chi_{B_\rho},$$

(if $|x| \neq \partial B_\rho$,) and

$$\Gamma_\rho \to \Gamma \quad \text{in } L^p \text{, with} \quad p < \frac{n}{n-2}.$$

2.4. $\triangle u$ **in the distribution sense.** Given $u \in L^1_{loc}$ we can define $\triangle u$ in the distribution sense: it generates a linear function on the space of smooth functions $\varphi \in C_0^\infty(\Omega)$ as follows:

$$\int (\triangle u)\varphi := \int u \, \triangle\varphi \; dx.$$

The left hand side is formal while the right hand side is well defined. An example of a distributional Laplacian we have

$$\triangle\Gamma = -\delta_0,$$

which can be established by passing $\rho \to 0$. In particular we will use the following definition for $\triangle u = f$ with f integrable:

Definition 2.1. Given u, $f \in L^1_{loc}(\Omega)$ we say that

$$\triangle u = f \quad \text{in } \Omega,$$

(in the weak sense) if

$$\int_\Omega u \, \triangle\varphi \; dx = \int_\Omega f \, \varphi \; dx, \qquad \forall \varphi \in C_0^2(\Omega).$$

An immediate consequence from the definition is
Stability: Assume that

$$\triangle u_n = f_n \quad \text{in } \Omega,$$

$$u_n \rightharpoonup u, \qquad f_n \rightharpoonup f.$$

Then

$$\triangle u = f.$$

Here $u_n \rightharpoonup u$ means that

$$\int_\Omega u_n \varphi \, dx \to \int_\Omega u \varphi \, dx, \qquad \forall \varphi \in C_0^\infty(\Omega).$$

2.5. **Weyl lemma.** *$\triangle u = 0$ in the weak sense in B_1 implies*

$$u \in C^\infty \qquad \text{and} \quad \|D^k u\|_{L^\infty(B_{1/2})} \le C(k)\|u\|_{L^1(B_1)}.$$

Idea: Prove the mean value theorem for u by using $\varphi = \Gamma_{\rho_1} - \Gamma_{\rho_2}$, then show that u is bounded, then Lipschitz and iterate (or first prove the result for the convolutions $u * \rho_\varepsilon$ where ρ_ε is a radial mollifier and let $\varepsilon \to 0$ at the end.)

2.6. **Scaling.** If $\triangle u = 0$ then $\tilde{u}(x) := \beta u(rx)$ satisfies $\triangle \tilde{u} = 0$.

A consequence is the Liouville theorem:

If $\triangle u = 0$ in $\mathbb{R}^n$ and $u(x) = O(|x|^m)$ at infinity for some m, then u is a polynomial of degree m.

2.7. **Subharmonic functions.** A function $u \in L^1_{loc}$ is subharmonic if

$$\triangle u \ge 0 \quad \text{in the distribution sense.}$$

This implies that $⨍_{B_r(x_0)} u dx$ are increasing with r hence u can be defined pointwise:

$$u(x_0) := \lim_{r \to 0} ⨍_{B_r(x_0)} u \, dx,$$

and then u is upper-semicontinuous i.e.

$$\limsup u(x_n) \le u(x_0) \quad \text{if } x_n \to x_0.$$

The following are equivalent:

a) $\triangle u \ge 0$,

b)

$$⨍_{B_r(x_0)} u \, dx \quad \text{is increasing in } r \text{ as long as } B_r(x_0) \subset \Omega.$$

Indeed a) $\Longrightarrow$ b) is as above. For b) $\Longrightarrow$ a) we notice that $u * \rho_\varepsilon$ still satisfies b) and then we get the conclusion by letting $\varepsilon \to 0$.

2.8. **Maximum principle.** If $u, v \in C^2(\Omega) \cap C(\overline{\Omega})$ and

$$\begin{cases} \triangle u \ge \triangle v & \text{in } \Omega, \\ u \le v & \text{on } \partial\Omega \end{cases} \qquad \Longrightarrow \qquad u \le v \quad \text{in } \Omega.$$

Indeed, we may assume $v = 0$, and then compare u with $t(|x|^2 - C)$ and let $t \to 0$. Important: it works for $Lu = a^{ij}(x)u_{ij} + b_i(x)u_i + c(x)u$ with $c \le 0$, with $A(x) = (a^{ij}(x)) > 0$.

Strong maximum principle: If $\triangle u \ge 0$ then $\max u$ cannot occur in Ω unless u is constant on that connected component.

2.9. **Harnack inequality.** (A quantified maximum principle)

If $u \ge 0$ and $\triangle u = 0$ in B_1 then $u \le Cu(0)$ in $B_{1/2}$.

It follows easily from the mean value property.

2.10. **Problems.**

1. Let $u \in C^2(B_1^+) \cap C^0(\overline{B}_1^+)$ be a harmonic function in the half ball
$$B_1^+ = B_1 \cap \{x_n > 0\}.$$

a) Show that if $u = 0$ on the flat portion of the boundary where $x_n = 0$, then its odd reflection with respect with $x_n = 0$ is harmonic in B_1.

b) Similarly, if $u_n = 0$ on the flat portion of the boundary, then its even reflection with respect with $x_n = 0$ is harmonic in B_1.

2. Let u be a harmonic function in B_1.

a) Find the optimal constant C in the gradient bound for harmonic functions
$$|\nabla u(0)| \le C \, \|u\|_{L^\infty(B_1)}.$$

b) Apply the gradient bound repeatedly in concentric balls to show that
$$|D^k u(0)| \le (Ck)^k \, \|u\|_{L^\infty(B_1)}$$
where $D^k u(0)$ denotes a derivative of u of order k along the coordinate directions.

c) Deduce that u is an analytic.

3. Show that if $\triangle u = f$ in Ω in the weak sense, then
$$\triangle(u * g_\varepsilon) = f * g_\varepsilon \quad \text{(in the weak sense) in} \quad \Omega_\varepsilon := \{x \in \Omega| \quad dist(x, \partial\Omega) > \varepsilon\},$$
for any $g_\varepsilon \in L^1$ that has compact support in B_ε.

4. a) Assume that $u \in C^2(\bar{\Omega})$ with $u = 0$ on $\partial\Omega \in C^1$. Let $\tilde{u}$ denote the extension of u by 0 outside Ω. Show that
$$\triangle \tilde{u} = \triangle u \ \chi_\Omega - u_\nu \, d\mathcal{H}^{n-1}|_{\partial\Omega},$$
where χ_Ω represents the characteristic function of Ω, and $d\mathcal{H}^{n-1}|_{\partial\Omega}$ denotes the surface measure of $\partial\Omega$.

b) If $\triangle u = 0$ in Ω and $u = u_\nu = 0$ on a potion of the boundary then u is identically 0.

5. Assume that $u \ge 0$ and $\triangle u \le 0$ in $\mathbb{R}^n$. Show that if $n = 2$ then u is constant, and if $n \ge 3$ then u is not necessarily a constant.

6. Assume that $u \in C^2 \cap L^\infty$ is a harmonic function defined in $B_1 \setminus \{0\}$. If u vanishes continuously on ∂B_1, then $u \equiv 0$.

7. (*Rellich identity*) Let $u \in C^2(B_r) \cap C^1(\overline{B_r})$ be a harmonic function. Compute
$$div\,\left(x|\nabla u|^2 - 2(x \cdot \nabla u)\nabla u\right),$$
and deduce that
$$(n-2)\int_{B_r} |\nabla u|^2 dx = \int_{\partial B_r} |\nabla u|^2 - 2u_\nu^2\, d\sigma.$$

8.(*Almgren frequency formula*). Let u be harmonic in B_1 and assume that it does not vanish identically. Then use the Rellich identity above to show that
$$\Phi(r) := \frac{r\int_{B_r} |\nabla u|^2 dx}{\int_{\partial B_r} u^2 d\sigma},$$
is monotone increasing in r. Moreover, $\Phi(r)$ is constant in r if and only if u is homogenous of degree $k \geq 0$.

9. Let u be harmonic in B_1. Show that
$$\int_{\partial B_1} u(ax)u(cx)dx = \int_{\partial B_1} u^2(bx)dx,$$
provided that $b^2 = ac$ and a, b, $c < 1$.

10. Find the radial solutions to the biharmonic equation
$$\triangle^2 u = 0,$$
and compute its fundamental solution.

11. Let $\alpha, \beta \in (0,1]$ and $u : [-2,2] \to [-1,1]$ be a 1D function that satisfies
$$\|u^h\|_{C^\beta([-1,1])} \leq 1, \quad \forall h \in [0,1/2],$$
where u^h denotes the discrete α-order discrete difference
$$u^h(t) := \frac{u(t+h) - u(t)}{h^\alpha}.$$

a) Show that if $\alpha = 1$ then $\|u\|_{C^{1,\beta}(I)} \leq C$, with $I = [-1/2, 1/2]$.

b) Show that if $\alpha + \beta > 1$ then $\|u\|_{C^{0,1}(I)} \leq C$.

c) Show that if $\alpha + \beta < 1$ then $\|u\|_{C^{\alpha+\beta}(I)} \leq C$.

For problem 8, we recall the definition of the Hölder norms $\|u\|_{C^\gamma(\Omega)}$ defined as
$$\|u\|_{C^\gamma} := \|u\|_{L^\infty} + \sup \frac{|u(x) - u(y)|}{|x-y|^\gamma},$$

$$\|u\|_{C^{1,\gamma}} := \|u\|_{L^\infty} + \|\nabla u\|_{C^\gamma}.$$

3. The Poisson equation

Next we solve the Dirichlet problem: Find $u \in C^2(\Omega) \cap C(\overline{\Omega})$ such that

$$\begin{cases} \triangle u = f & \text{in } \Omega, \\ u = \varphi & \text{on } \partial\Omega. \end{cases}$$

Solution is unique by the maximum principle.

We denote by $\Gamma_{x_0}(x) = \Gamma(x - x_0)$ the fundamental solution with pole at x_0, and we have

$$\int_\Omega \triangle u \ \Gamma_{x_0} dx = \int_\Omega u \ \triangle \Gamma_{x_0} dx + \int_{\partial\Omega} u_\nu \Gamma_{x_0} - u \ (\Gamma_{x_0})_\nu \, d\sigma,$$

or

$$u(x_0) = -\int_\Omega f \ \Gamma_{x_0} dx + \int_{\partial\Omega} u_\nu \Gamma_{x_0} - u \ (\Gamma_{x_0})_\nu \, d\sigma.$$

3.1. **Green's function.** Denote by

$$G_{x_0} := \Gamma_{x_0} - h_{x_0},$$

with h_{x_0} the harmonic function such that

$$G_{x_0} = 0 \quad \text{on } \partial\Omega, \quad \triangle G_{x_0} = -\delta_{x_0}.$$

Then we have the representation formula

$$u(x_0) = -\int_\Omega f \ G_{x_0} \, dx - \int_{\partial\Omega} \varphi \ (G_{x_0})_\nu \, d\sigma.$$

Exercise:

1) If $G(x, y) := G_y(x)$, then $G(x, y) = G(y, x)$ - by writing the representation formula for $G(x, x_1)$.

(Thus $h(x, x_0) := h_{x_0}(x)$ is symmetric and harmonic in both variables.)

2) If $f \in L^1$ then

$$\triangle (f * \Gamma) = -f.$$

This can be seen from the stability by approximating with $f * \Gamma_\rho$ and letting $\rho \to 0$.

If $f \in C^1$ then $f * \Gamma \in C^2$ and the equality above is in the classical sense.

Explicit Green functions:

1) $\Omega = \mathbb{R}^+_n$, let $\bar{x_0} = x_0 - 2x_0 \cdot e_n$,

$$G_{x_0} = \Gamma_{x_0} - \Gamma_{\bar{x_0}}, \qquad \nabla G_{x_0} \cdot e_n = c'_n \ \frac{x_0 \cdot e_n}{|x - x_0|^n}.$$

2) $\Omega = B_1$, let $x_0^* := x_0 |x_0|^{-2}$,

$$G_{x_0} := \Gamma_{x_0} - |x_0|^{2-n} \Gamma_{x_0^*}, \qquad -(G_{x_0})_\nu = c'_n \ \frac{1 - |x_0|^2}{|x - x_0|^n}.$$

3.2. **Dirichlet problem on a ball.**

Theorem 3.1.

$$u(x) = \int_{\partial B_1} c'_n \ \frac{1 - |x|^2}{|x - y|^n} \ \varphi(y) \, d\sigma,$$

solves the Dirichlet problem in B_1 (with $f = 0$).

It follows from the 3 properties of the Poisson kernel

$$P(x,y) := c_n' \frac{1-|x|^2}{|x-y|^n}, \qquad y \in \partial B_1, x \in B_1,$$

$$\triangle_x P = 0, \quad P \geq 0, \quad \int_{\partial B_1} P d\sigma = 1, \quad \text{and} \quad \int_{\partial B_1 \cap \{|y-x|>\delta\}} P d\sigma \to 0 \quad \text{as } |x| \to 1.$$

3.3. **Subharmonic functions.** The following are equivalent for an uppersemicontinuous function u:

1) u is subharmonic
2) u cannot be touched by above at an interior point by $v \in C^2$ with $\triangle v < 0$.
3) u cannot be touched by above at an interior point by a quadratic polynomial P with $\triangle P < 0$.

Indeed 1) $\Longrightarrow$ 2) by the strong maximum principle. For 2) $\Longrightarrow$1) we first show that

$$\fint_{\partial B_r(x_0)} u \, d\sigma \quad \text{is nondecreasing in } r$$

by comparing u with a harmonic function which is above u on $\partial B_r(x_0)$. Then the results follows by integration in r. The fact that 2) and 3) are equivalent is obvious.

Consequence: If u is subharmonic and continuous and we replace it in a ball $B_r(x_0) \subset \Omega$ by the harmonic function with the same boundary data on $\partial B_r(x_0)$, then we still have a subharmonic function which is above u.

3.4. **Perron's method.**

Theorem 3.2. *Assume that Ω is a bounded domain that satisfies an exterior ball condition. Then the Dirichlet problem is solvable by*

$$u = \sup\{v| \quad v \in C(\overline{\Omega}) \quad \textit{subharmonic, } v \leq \varphi \textit{ on } \partial\Omega\}.$$

Step 1: use barriers to show that the set of subsolutions is non-empty and obtain $\underline{\varphi} \leq u \leq \bar{\varphi}$.

Step 2: show that u is harmonic by using $u \geq h$ in $B_r(x_0)$ with $u(x_0) = h(x_0)$, h harmonic. In fact $u = h$ since for each $x_1 \in B_r(x_0)$ we can find another harmonic $h' \geq h$ such that $u(x_1) = h'(x_1)$, $h'(x_0) = h(x_0)$.

Conclusion: We can always solve the Dirichlet problem (in the classical sense) uniquely if Ω has the exterior ball condition and $f \in C^1(\bar{\Omega})$.

If $f \in L^p(\Omega)$ with $p > \frac{n}{2}$ then we can solve the problem with $u \in C(\overline{\Omega})$ so that $\triangle u = f$ holds in the distribution sense (we reduce it to $f = 0$ by adding $f * \Gamma$).

3.5. **Problems.**

1. a) Assume that $u \in C^2(\Omega)$ solves

$$Lu = trA(x)D^2u + b(x) \cdot \nabla u + c(x)u = 0,$$

with $\lambda I \leq A(x) \leq \Lambda I$, $|b| \leq K$ for some positive constants λ, Λ, K. Construct an explicit function $\Psi \in C^2$ such that

$$\Psi \geq 0, \quad L\Psi < 0 \quad \text{in } \bar{\Omega},$$

provided that $\Omega \subset \{|x_n| \le d\}$ for some constant d, and $c(x) \le c_0$ with c_0 a small constant depending on $\lambda, \Lambda, n, K, d, n$.

b) Show that if in addition we assume that Ω is bounded, then the maximum principle holds:
If $u, v \in C^2(\Omega) \cap C(\bar\Omega)$ satisfy $Lu \ge Lv$ and $u \le v$ on $\partial\Omega$, then $u \le v$ in Ω.
Give a counterexample to this statement when Ω is unbounded.

c) Using a scaling argument show that $c_0 \to \infty$ as $d \to 0$. Conclude that the maximum principle holds for L in sufficiently small balls of radius depending on λ, Λ, n, $\|b\|_{L^\infty}$, $\|c\|_{L^\infty}$.

2. *Hopf lemma*
a) Construct a radial function $\Phi \ge 0$ in $\Phi \in C^2(\bar B_1 \setminus B_{1/2})$ such that $\Phi = 0$ on ∂B_1, $\Phi = 1$ on $\partial B_{1/2}$, $|\nabla \Phi| > 0$ on ∂B_1, and

$$tr A(x) D^2 \Phi > 0,$$

for any matrix $A(x)$ with $\lambda I \le A(x) \le \Lambda I$.

b) Assume that L is as in Problem 1 above with $\|b\|_{L^\infty}$, $\|c\|_{L^\infty}$ sufficiently small. By continuity $L\Phi > 0$, and conclude that if $Lu \le 0$ and $u \ge 0$ in B_1, and $u > 0$ on $\partial B_{1/2}$ then $u > 0$ in B_1.

c) Assume that $u \ge 0$ is continuous in $\overline{B}_1$, $Lu \le 0$ and $u(x_0) = 0$ for some $x_0 \in \partial B_1$. Then

$$u_\nu(x_0) < 0.$$

(here $\|b\|_{L^\infty}$, $\|c\|_{L^\infty}$ are bounded, but not necessarily small.)

4. a) If $\Omega_1 \subset \Omega_2$ are bounded domains, show that $G_{\Omega_1} \le G_{\Omega_2}$ in their common domain of definition.

b) Assume that Ω is a bounded C^2 domain and $0 \in \partial\Omega$. Show that as $y \to 0$, $G(x, y) \to 0$ uniformly in x outside any neighborhood of 0.

c) Conclude that also $G_\nu(x, y) \to 0$ uniformly in $x \in \partial\Omega$ outside any neighborhood of 0, hence the Poisson kernels $-G_\nu(x, y)$ for the domain Ω approximate δ_0 as $x \to 0$.

5 *Fatou's theorem*
Assume that u is a bounded harmonic function in B_1. We can define the boundary values for u on ∂B_1 by showing that there exists an L^∞ function φ defined on the ∂B_1 such that

$$u(x) = \int_{\partial B_1} P(x, y) \varphi(y) dy,$$

where $P(x, y)$ is the Poisson kernel for the ball.

Moreover show that for a.e. $y \in \partial B_1$ we have $u(ry) \to \varphi(y)$ as $r \to 1^-$.
(Hint: Find φ as a weak limit of $u(ry)$ as $r \to 1^-$.)

Remarks: If u is only bounded below then φdy is replaced by a measure. Also one can check that $u(x) \to \varphi(y)$ as $x \to y$ nontangentially meaning that x belongs to a cone with vertex at y, axis $-y$, and angle $\theta < \pi/2$.

Can you think of an example of a harmonic function in B_1 which is not bounded either from below or above?

6. *Mean Value theorem for the heat equation*

a) Show that

$$\Gamma(x,t) := c_n t^{-n/2} e^{-\frac{|x|^2}{4t}}, \quad \text{if } t > 0,$$

and $\Gamma = 0$ if $t \leq 0$ is the "fundamental solution" to the heat eqaution in the sense that the equation

$$\triangle_x \Gamma - \Gamma_t = -\,\delta_{(0,0)}$$

holds in the distribution sense for an appropriate constant c_n.

b) Let $h(s) = s$ if $s < 1$, $h(s) = 1 + \log s$ if $s \geq 1$ (so that $h'' s^2 = -1$). Show that

$$v(x,t) := h(\Gamma(x,-t))$$

solves

$$\triangle\, v + v_t = -c_n' \,\frac{|x|^2}{t^2}\, \chi_E,$$

where $E \subset \mathbb{R}^{n+1}$ is in the reflection of the set $\Gamma > 1$ with respect to $t = 0$. Notice that the right hand side integrates to -1.

c) Show that if $u \in C^2$ solves the heat equation $\triangle u - u_t = 0$ in a domain containing E then u can be written as a weighted average of the values of u in E

$$u(0,0) = c_n' \int_E u(x,t)\, \frac{|x|^2}{t^2}\, dxdt.$$

d) Use the scaling of the equation to write $u(x,t)$ in terms of averages of u in the sets $(x,t) + E_r$, with E_r rescalings of E. (E_r is the "heat ball" of size r).

7. *Green function for the heat equation*

a) Assume that u solves the heat equation in the cylinder $Q = \Omega \times [0,T]$ and $u = \varphi$ on the parabolic boundary

$$\partial_p Q := (\Omega \times |\{0\}) \cup \ (\partial\Omega \times [0,T]).$$

Define the Green function $G_{(x_0,t_0)}$ similarly as we did for the Laplace equation and express $u(x_0,t_0)$ in terms of φ and $G_{(x_0,t_0)}$ (you may assume the solvability of the Dirichlet problem for the heat equation).

b) When $\Omega = [0,1]^n$ is the unit cube, write the Green function as a series involving the fundamental solutions (do it for $n = 1$ for simplicity).

The next problem discusses the Perron's method in arbitrary bounded domains, and the uniqueness of the Dirichlet problem for continuous boundary data.

8. *Perron-Wiener-Brélot Theorem*

Let Ω be a bounded domain, and for a bounded continuous function $u : \Omega \to \mathbb{R}$ denote by $\limsup u$, $\liminf u$ its upper (lower) envelope extensions to $\partial\Omega$. Let φ be a bounded function on $\partial\Omega$. Denote by

$$\underline{\mathcal{A}}_\varphi := \{u : \Omega \to \mathbb{R}, \text{ continuous, subharmonic, } \limsup u \leq \varphi\},$$

$$\underline{H}_\varphi := \sup u, \qquad \text{among all } u \in \underline{\mathcal{A}}_\varphi,$$

and similarly, we define $\overline{H}_\varphi$.

a) Show that $\underline{H}_\varphi \leq \overline{H}_\varphi$ and both functions are harmonic in Ω.

b) If $\underline{H}_\varphi$ and $\overline{H}_\varphi$ do not coincide then there exists a linear function l and a point $x_0 \in \partial\Omega$ such that

$$w := l - (\overline{H}_\varphi - \underline{H}_\varphi) > 0 \quad \text{in } \Omega,$$

and

$$l(x_0) > 0, \quad \liminf w(x_0) = 0, \quad \liminf w(x) > 0 \quad \forall x \in \overline{\Omega} \setminus \{x_0\}.$$

c) If φ is continuous then show that $\underline{H}_\varphi = \overline{H}_\varphi$.

Otherwise, let $y_n \in \Omega$, $y_n \to x_0$ with $w(y_n) \to 0$, and use w as a barrier at x_0 to show $\overline{H}_\varphi(y_n) \to \varphi(x_0)$, $\underline{H}_\varphi(y_n) \to \varphi(x_0)$ which contradicts $l(x_0) > 0$.

d) (*Harmonic measure*) Deduce that there exists a Borel measure μ_x such that

$$H_\varphi(x) := \int_{\partial\Omega} \varphi \, d\mu_x, \qquad x \in \Omega, \quad \varphi \in C(\partial\Omega),$$

where H_φ is the unique harmonic function from c).

4. Potential theory

If $\triangle u = f$ in B_1, then we study the properties of u by looking at the Newtonian potential (or Riesz potential) of f:

$$w := f * \Gamma,$$

and use that $u = -w + h$ with h a harmonic function.

If $f \in L^p$ with $p > \frac{n}{2}$ then w is a continuous function.

If f is bounded then $w \in C^1$ and $w_i = f * \Gamma_i$.

Lemma 4.1. *If $f \in C^\alpha$ then $w \in C^2$ and*

$$w_{ij} = -\frac{\delta_i^j}{n} f + pv\, f * \Gamma_{ij}.$$

Indeed $w_\rho := f * \Gamma_\rho$ satisfies

$$\partial_{ij} w_\rho = -\frac{\delta_i^j}{n} f * \frac{\chi_{B_\rho}}{|B_\rho|} + f * \left(\Gamma_{ij}\chi_{B_\rho^c}\right).$$

As $\rho \to 0$ the right hand side converges uniformly and we obtain the result.

Next we study the properties of the convolution

$$pv\, f * \Gamma_{ij}.$$

We check the Fourier transform of this convolution. If $u \in C_0^\infty$ and $\triangle u = f$, then $u = -w$. We use the Fourier transform $u = \int \hat{u}(\xi) e^{ix\cdot\xi} d\xi$ and find

$$-|\xi|^2 \hat{u} = \hat{f}, \quad \hat{u}_{ij} = \frac{\xi_i \xi_j}{|\xi|^2}\, \hat{f},$$

hence formally

$$\hat{\Gamma}_{ij} = -\frac{\xi_i \xi_j}{|\xi|^2}, \quad \text{if } i \neq j, \text{ and } \quad \hat{\Gamma}_{ii} = \frac{1}{n} - \frac{\xi_i^2}{|\xi|^2}.$$

4.1. Singular integrals.

Let K be homogenous of degree $-n$ function, with

$$|\nabla K| \leq \frac{C}{|x|^{n+1}}, \qquad \int_{\partial B_1} K\, d\sigma = 0.$$

Definition 4.2. We define the singular operator Tf as

$$Tf := pv f * K,$$

which is well defined if $f \in C_0^\alpha$.

Exercise: Check that $\hat{Tf} = \hat{K}\, \hat{f}$ for some $\hat{K}$ homogenous of degree 0.

T is an operator of "order 0" - it commutes with dilations (and translations).

Proposition 4.3. *Let $\alpha \in (0,1)$. Then*

$$[Tf]_{C^\alpha} \leq C(K)[f]_{C^\alpha},$$

and if $\operatorname{supp} f \subset B_R$ then

$$\|Tf\|_{C^\alpha} \leq C(K,R)\|f\|_{C^\alpha},$$

Assume for simplicity $[f]_{C^\alpha} \le 1$. Let $K_z := K(z-x)$ denote the kernel with pole at z and then, by integrating first on spheres centered at z we find

$$Tf(z) = \int_{B^c} (f(x)-f(z))K_z \, dx + O(1)a^\alpha \quad \text{if } B \subset B_{4a}(z).$$

By taking $a = |y-z|$, and B the ball of radius $2a$ centered at $\frac{1}{2}(y+z)$,

$$Tf(z) - Tf(y) = \int_{B^c} (f(x)-f(z))(K_z - K_y) + (f(y)-f(z))K_y \, dx + O(1)a^\alpha.$$

Since K_y integrates to 0 on spheres centered at y we find

$$\int_{B^c} (f(y)-f(z))K_y \, dx = O(1)a^\alpha.$$

Also, by integrating on spheres centered at z

$$\left| \int_{B^c} (f(x)-f(z))(K_z - K_y) \, dx \right| \le C \int_a^\infty r^\alpha \, a r^{-n-1} r^{n-1} dr = O(1)a^\alpha.$$

Proposition 4.4 (Boundary version)**.** *Assume that $f \in C^\alpha(\mathbb{R}^+_n)$ is extended odd with respect to $x_n = 0$, and K is either even in the x_n variable or odd in a tangential x_i direction with $i < n$. Then the estimates above hold for Tf restricted to $\mathbb{R}^n_+$.*

If we write the integration only in $\mathbb{R}^+_n$ we have

$$Tf(z) = \int_{\mathbb{R}^+_n} (f(x)-f(z))(K_z - \bar{K}_{\bar{z}}) \, dx + c_0(K) f(z),$$

with

$$c_0(K) = pv \int_{\mathbb{R}^n_+} K_z - \bar{K}_{\bar{z}} \, dx = \int_{\mathbb{R}^n} K_z \, (\chi_{\{x_n>0\}} - \chi_{\{x_n<0\}}) \, dx,$$

which is independent of z and well defined since

$$\int_{\partial B_1} K \, (\chi_{\{x_n>0\}} - \chi_{\{x_n<0\}}) d\sigma = 0.$$

The rest of the proof is the same.

4.2. **$C^{2,\alpha}$ estimates for Laplace equation.** The propositions above give the following $C^{2,\alpha}$ estimates for solutions to Laplace equation.

Proposition 4.5. *Assume $f \in C^\alpha(B_1)$ and*

$$\begin{cases} \triangle u = f & \text{in } B_1, \\ u = \varphi & \text{on } \partial B_1. \end{cases}$$

Then $u \in C^{2,\alpha}(B_1)$ and

$$\|u\|_{C^{2,\alpha}(B_{1/2})} \le C(n,\alpha) \, \left(\|\varphi\|_{L^\infty} + \|f\|_{C^\alpha} \right).$$

Proposition 4.6 (Boundary version)**.** *Assume $f \in C^\alpha(B^+_1)$ and $\varphi \in C^{2,\alpha}$ near $\{x_n = 0\}$.*

$$\begin{cases} \triangle u = f & \text{in } B^+_1, \\ u = \varphi & \text{on } \partial B^+_1. \end{cases}$$

Then $u \in C^{2,\alpha}(B^+_1)$ and

$$\|u\|_{C^{2,\alpha}(B^+_{1/2})} \le C(n,\alpha) \, \left(\|u\|_{L^\infty} + \|\varphi\|_{C^{2,\alpha}} + \|f\|_{C^\alpha} \right).$$

Remark 4.7. By using the Kelvin transform, we can transform locally ∂B_1 into a flat boundary without changing the Laplace operator. Thus we also have the global estimate version of Proposition 4.5

$$\|u\|_{C^{2,\alpha}(\overline{B}_1} \leq C(n,\alpha)\ \left(\|\varphi\|_{C^{2,\alpha}(\partial B_1)} + \|f\|_{C^{\alpha}(\overline{B}_1)}\right).$$

4.3. **An alternate approach.** (Campanato method)

Definition 4.8. Let $\alpha \in (0,1]$. We say that u is pointwise $C^{k,\alpha}$ at x_0 if there exists a polynomial P of degree k such that

$$|u - P| \leq C|x - x_0|^{k+\alpha}.$$

We say that P is the k-th order tangent polynomial to u at x_0.

Exercise: Show that it suffices to require in the definition above that for each $r = 2^{-k}$, $k \geq 0$, there exists a polynomial P_r (depending on r) of degree k such that

$$|u - P_r| \leq Cr^{k+\alpha}.$$

Characterization of $C^{k,\alpha}$ spaces:

Assume that $\alpha \in (0,1]$ and Ω a Lipschitz domain.

Then $u \in C^{k,\alpha}(\overline{\Omega})$ is equivalent to say that u is pointwise $C^{k,\alpha}$ at all points $x \in \overline{\Omega}$ with uniform constant C for all x.

For example $u \in C^{\alpha}(\overline{\Omega})$ if and only if there exist r_0 and M such that

$$\forall\, x \in \overline{\Omega},\ r \leq r_0, \quad \exists\, a_{x,r} \in \mathbb{R} \quad \text{such that} \quad |u - a_{x,r}| \leq Mr^{\alpha} \quad \text{in} \quad B_r(x) \cap \Omega.$$

Similarly for $u \in C^{k,\alpha}$ we write instead

$$\exists\, P_{x,r} \quad \text{polynomial of degree } k \quad \text{such that} \quad |u - P_{x,r}| \leq Mr^{k+\alpha} \quad \text{in} \quad B_r(x) \cap \Omega.$$

Proof of Proposition 4.5. We show that u is pointwise $C^{2,\alpha}$ at the origin. After dividing by an appropriate constant we may assume that $\|f\|_{C^{\alpha}}, \|u\|_{L^{\infty}} \leq \delta$ small, and we show inductively on radii $r = \rho^k$ with ρ small, $k = 0, 1, 2, ..$ that

$$|u - P_r| \leq r^{2+\alpha}, \qquad \text{with} \quad \triangle P_r = f(0).$$

The case $k = 0$ is obvious: we choose $P = \frac{f(0)}{2n}|x|^2$.

Assume the induction hypothesis holds for $r = \rho^k$, and we want to prove it for ρr as well. Let

$$\tilde{u}(x) := \frac{(u - P_r)(rx)}{r^{2+\alpha}},$$

denote the rescaled error, and then in B_1 we have

$$|\triangle \tilde{u}| \leq \delta, \quad |u| \leq 1.$$

We compare w with the harmonic function with the same boundary data as $\tilde{u}$ on ∂B_1 and obtain

$$\|\tilde{u} - w\|_{L^{\infty}} \leq \delta, \quad |w - P_w| \leq C\rho^3 \quad \text{in} \quad B_{\rho},$$

with P_w the quadratic part of w at 0, and C a constant depending only on n. We choose δ and ρ such that

$$\|\tilde{u} - P_w\| \leq \rho^{2+\alpha},$$

which gives the desired conclusion. $\square$

Exercise: Prove the boundary version for domains with $\partial\Omega \in C^{2,\alpha}$.

4.4. **Problems.**

1. a) Let $K(x) := \max\{0, 1 - |x|^{2-n}\}$ be defined in B_{2M}, and notice that K vanishes in B_1 and it is (super)harmonic in $B_{2M} \setminus B_1$. Let K_r be the family of rescaligns of K,

$$K_r(x) := r^\alpha K((x - y_r)/r), \quad \text{with} \quad y_r = M(1-r)e_1,$$

which are defined in the corresponding rescaled domains (notice that $K_1 = K$.)

Show that if α is chosen sufficiently small depending on n and M, then

$$\inf_{r\in[0,1]} K_r$$

is a barrier for the piece of cone $\mathcal{C} = \cup B_r(y_r)$ meaning that it is continous, it vanishes on $\mathcal{C}$ and it is superharmonic and positive in $B_{2M} \setminus \mathcal{C}$.

b) Can you sketch a similar argument to find a barrier outside the $n-1$ dimensional cone $\mathcal{C} \cap \{x_n = 0\}$ by first constructing a barrier for the $n-1$ dimensional ball $B_1 \cap \{x_n = 0\}$? (start with $n = 2$.)

2. Let $\triangle u = f$ in $\Omega \subset \mathbb{R}^n$. Show that the Kelvin transform of u

$$u^*(x) := |x|^{2-n} u\left(\frac{x}{|x|^2}\right)$$

satisfies

$$\triangle u^*(x) = |x|^{-n-2} f\left(\frac{x}{|x|^2}\right).$$

3. a) Let Ω be a Lipschitz domain. Show that if for any ball $B_r(x)$ with $x \in \overline{\Omega}$ there exists a polynomial $P_{x,r}$ of degree k such that

$$\|u - P_{x,r}\| \le M r^{k+\alpha} \quad \text{in } B_r(x) \cap \overline{\Omega},$$

then $u \in C^{k,\alpha}(\overline{\Omega})$ and

$$[u]_{C^{k,\alpha}} \le CM.$$

Above $\alpha \in (0,1]$ and $k \ge 0$ is an integer. (It suffices to write the proof with $k = 1$ for simplicity).

4. a) Assume that $\triangle u = f$ in B_1 and $\|u\|_{L^\infty}, \|f\|_{L^\infty} \le 1$. Using the Campanato iteration approach show that

$$|u - P_{x,r}| \le Cr^2 \quad \text{in } B_r(x) \subset B_{1/2},$$

for some harmonic quadratic polynomial $P_{x,r}$, and a constant C depending only on the dimension. Conclude that u is "almost $C^{1,1}$" apart from a logharithmic correction:

$$|u(x) - u(0) - \nabla u(0) \cdot x| \le C|x|^2 |\log |x||,$$

and $|\nabla u(x) - \nabla u(y)| \le C|x-y||\log|x-y||$ for $x, y \in B_{1/2}$.

b) Show that the estimates above are optimal if, say in 2D, $f = 1$ in the first and third quadrants and $f = -1$ in the second and fourth quadrants.

On the other hand, if the discontinuity of f occurs only along one direction, say $f = 1$ in $\{x_1 > 0\}$ and $f = -1$ in $\{x_1 < 0\}$ then $u \in C^{1,1}$.

5. Let $f \in C^\alpha$ be a periodic function of period 1 in the coordinate direction (or we can think of f defiened on the flat torus $T = [0,1)^n$). Solve by Perron's method the equation

$$\triangle u - \varepsilon u = f$$

with u periodic, and denote by u_ε the unique solution. Show that as $\varepsilon \to 0$, $u_\varepsilon - u_\varepsilon(0)$ is uniformly Lipschitz and it converges (up to subsequences) to a solution $\bar{u}$ of the equation

$$\triangle \bar{u} = f - \int_T f \, dx.$$

6. Assume that u is a positive harmonic function in $\Omega := B_1 \setminus \mathcal{C}_\beta$ where C_β is the cusp

$$C_\beta := \{x_n + |x'|^\beta \le 0\}, \quad \beta \in (0,1),$$

and assume that u vanishes continuously on $\partial C_\beta \setminus \{0\}$. Show that u is discontinous at the origin if $n \ge 4$.

(Hint: look at $\min_{D_r} u$ where D_r are the cylinders $D_r = \{|x'| = r, \quad |x_n| \le r^\gamma\}$ for some $\gamma \in (\beta, 1)$ for the sequence $r = 2^{-k}$.)

Does the same result hold in dimension $n = 3$?

5. Schauder estimates

Theorem 5.1 (Schauder estimates)**.** *Assume that $u \in C^2$ solves*

$$a^{ij}u_{ij} + b_i u_i + cu = f \quad in\ B_1. \tag{5.1}$$

with $A = (a^{ij}) \geq \lambda I$, and $a^{ij}, b_i, c, f \in C^\alpha$. Then $u \in C^{2,\alpha}(B_1)$ and

$$\|u\|_{C^{2,\alpha}(B^+_{1/2})} \leq C\ \left(\|u\|_{L^\infty} + \|f\|_{C^\alpha}\right),$$

with C depending on the data.

Theorem 5.2 (Boundary version)**.** *Assume that $\Omega = \{x_n > g(x')\}$ with $g \in C^{2,\alpha}$, $g(0) = 0$ and $u \in C^2(\Omega \cap B_1)$, continuous up to $\partial\Omega$ solves*

$$a^{ij}u_{ij} + b_i u_i + cu = f \quad in\ \Omega \cap B_1.$$

with $a^{ij}, b_i, c, f \in C^\alpha$ as above. If $u = \varphi \in C^{2,\alpha}$ on $\partial\Omega \cap B_1$, then $u \in C^{2,\alpha}(B_1)$,

$$\|u\|_{C^{2,\alpha}(\Omega \cap B_{1/2})} \leq C\ \left(\|u\|_{L^\infty} + \|\varphi\|_{C^{2,\alpha}} + \|f\|_{C^\alpha}\right),$$

with C depending on the data.

5.1. **Apriori estimates.**

Lemma 5.3. *Assume that $u \in C^{2,\alpha}(B_1)$ and satisfies (5.1) with*

$$\|a^{ij} - \delta_i^j\|_{C^\alpha}, \|b_i\|_{C^\alpha}, \|c\|_{C^\alpha} \leq \delta,$$

for some $\delta > 0$ universal. Then

$$[u]_{C^{2,\alpha}(B_{1/2})} \leq C\left(\|u\|_{L^\infty} + \|f\|_{C^\alpha}\right).$$

Step 1: We show that

$$[u]_{C^{2,\alpha}(B_{1/2})} \leq C\left(\|u\|_{L^\infty} + \|f\|_{C^\alpha}\right) + \mu \max_{B_{1/8}(x) \subset B_{3/4}} [u]_{C^{2,\alpha}(B_{1/8}(x))},$$

with $\mu = 2^{-7}$.

Indeed,

$$\triangle u = f + (\delta_i^j - a^{ij})u_{ij} - b^i u_i - cu,$$

and then we use the interpolation inequality $\|u_{ij}\|_{L^\infty} \leq C\|u\|_{L^\infty} + [u]_{C^{2,\alpha}}$.

Corollary: After rescaling $\tilde{u}(x) = u(rx)$ with $r \leq 1$, then

$$[u]_{C^{2,\alpha}(B_{r/2})} \leq Cr^{-3}\left(\|u\|_{L^\infty(B_r)} + \|f\|_{C^\alpha(B_r)}\right) + \mu\ \max\ [u]_{C^{2,\alpha}(B_{r/8}(x))}.$$

Step 2: If

$$a_k \leq C_1 2^{Mk} + \mu a_{k+1},$$

with $\mu < 2^M$. Then $a_k \to \infty$ if $a_0 > C(C_1, M)$ large. (It follows by induction).

Conclusion: Theorems 5.1 and 5.2 hold under the assumptions $u \in C^{2,\alpha}(B_1)$ respectively $u \in C^{2,\alpha}(\overline{B}_1^+)$. These assumptions are removed in the next subsection.

5.2. **Existence.** We assume that $c \leq 0$, or $c \leq \delta(\lambda, \Lambda, \Omega)$ so that the maximum principle holds.

Theorem 5.4. *Assume that $\partial\Omega \in C^{2,\alpha}$, $f \in C^\alpha$, $\varphi \in C^{2,\alpha}$. There exists a unique solution $u \in C^{2,\alpha}(\overline{\Omega})$ which solves the Dirichlet problem*

$$Lu = f \quad \text{in } \Omega, \quad u = \varphi \quad \text{on } \partial\Omega,$$

and

$$\|u\|_{C^{2,\alpha}(\overline{\Omega})} \leq C(\|f\|_{C^\alpha(\overline{\Omega})} + \|\varphi\|_{C^{2,\alpha}(\partial\Omega)}).$$

By a solution to the Dirichlet problem we mean a function $u \in C^2(\Omega) \cap C^0(\overline{\Omega})$. The statement above says that u is in fact $C^{2,\alpha}$ up to the boundary if $\varphi \in C^{2,\alpha}$.

After subtracting a $C^{2,\alpha}$ extension of the boundary data φ from u, we may assume $\varphi = 0$.

Functional analysis fact (continuity method for linear operators):

If $L_t : C_0^{2,\alpha} \to C^\alpha$ are linear maps continuous in $t \in [0, 1]$, such that L_0 is invertible and

$$L_t x = y \quad \Longrightarrow \quad \|x\| \leq C\|y\|,$$

then, all L_t are invertible, with $\|L_t^{-1}\| \leq C$.

Indeed, we use

If $A, B : X \to Y$ are bounded linear maps between Banach spaces, and A is invertible and $\|B\| < \|A^{-1}\|^{-1}$ then $A + B$ is invertible as well.

Then we can find a finite sequence $0 < t_1 < t_2 .. < 1$ with $\|L_{t_{k+1}} - L_{t_k}\| < C^{-1}$.

In our case

$$L_t = (1 - t)\triangle + tL.$$

I) From the hypotheses on c, (and $\varphi = 0$) we find $\|u\|_{L^\infty} \leq C\|f\|_{L^\infty}$. We can apply the result above in the case when $\Omega = B_1$, since by the Kelvin transform we can prove the invertibility for $L_0 = \triangle$ (see Remark 4.7). Thus Theorem 5.4 holds for $\Omega = B_1$.

II) In the case when φ is only continuous on ∂B_1, we can solve the Dirichlet problem by approximation. Let $\varphi_k \in C^{2,\alpha}$ converge uniformly to φ. Then by maximum principle the corresponding solutions $u_k \in C^{2,\alpha}(\overline{B}_1)$ converge uniformly to u. From the interior Schauder estimates for u_k's we conclude that $u \in C^{2,\alpha}(B_1) \cap C^0(\overline{B}_1)$. This means that C^2 solutions to general equations $Lu = f$ with C^α Holder coefficients and data are in fact $C^{2,\alpha}$ locally (after a dilation in a small ball, they satisfy the smallness hypothesis for c.)

III) Using domain variations we can prove Theorem 5.4 for domains Ω which are $C^{2,\alpha}$ diffeomorphisms of B_1, by continuity method. Indeed, a diffeomorphism of a solution u solves an equation of the same type since

$$u(x) = v(\Phi(x)) \quad \Longrightarrow \quad Du = DvD\Phi, \quad D^2u = (D\Phi)^T D^2 v D\Phi + DvD^2\Phi.$$

IV) As above, by approximation (and the apriori boundary Schauder estimates) we can show that if $\varphi \in C^{2,\alpha}$ near a point on $\partial\Omega \in C^{2,\alpha}$, then a solution $u \in C^2(\Omega) \cap C^0(\partial\Omega)$ is in fact $C^{2,\alpha}$ up to the boundary in a neighborhood of that point. Now Theorem 5.4 holds for general domains Ω.

5.3. **Fredholm alternative.** Let L be as above. Then $L : C_0^{2,\alpha} \to C^\alpha$ is an isomorphism and its inverse is a compact operator

$$Lu = f \quad \Longrightarrow \quad u = Tf, \quad T = L^{-1} \quad \text{compact operator form } C^\alpha \text{ to itself.}$$

If c does not necessarily have a sign the the equation can be written as

$$Lu + \lambda u = f \quad \Longleftrightarrow \quad (T + \lambda^{-1} I)u = Tf,$$

which is invertible except for a sequence of $\lambda_k \to \infty$.

The main point is that when $T : X \to X$ is compact, its range is almost finite dimensional in the sense that $T(B_1^X)$ is included in an ε neighborhood of a finite dimensional subspace. It follows that the analysis for the range and kernel of $T + \lambda^{-1} I$ is similar to the finite dimensional case: it is invertible if and only if the homogenous equation has a unique solution $u = 0$.

Example: Eigenvalues for the Laplace equation

$$\triangle u + \lambda u = 0, \quad u = 0 \quad \text{on } \partial\Omega.$$

We can compute the λ's explicitly when Ω is a box $\Pi\,[a_i, b_i]$.

For the inhomogeneous equation the right hand side f must be orthogonal (in L^2) to the λ eigenspace in order to have existence.

Remark: In general we can guarantee existence to an equation $Lu = f$ as long as we can bound $\|u\|_{L^\infty}$.

5.4. **Existence for continuous data.** We can rewrite Theorem 5.4 for data φ continuous and domains Ω that satisfy the exterior cone condition with

$$u \in C^{2,\alpha}(\Omega) \cap C(\overline{\Omega}).$$

This can be done either by approximation with $C^{2,\alpha}$ data or by Perron's method.

5.5. **Higher regularity.** Theorems 5.1 and 5.2 hold as we have $a^{ij}, b^i, c, \partial\Omega \in C^{k,\alpha}$, and $u \in C^{k+2,\alpha}$.

This becomes obvious as we differentiate the equation one time and apply induction. For the first time we need to work with the discrete differences

$$u_e^h(x) := \frac{u(x + he) - u(x)}{h}.$$

For the boundary version the u_{nnn} can be recovered from the other derivatives.

5.6. **Other boundary conditions.** We can do the same for Neumann boundary condition $u_\nu = 0$ on $\partial\Omega$ or mixed boundary conditions $b \cdot \nabla u + du = \varphi$ on $\partial\Omega$ with $b \cdot \nu > 0$. For this we first need to develop the regularity in B_1^+ (exercise).

5.7. **Problems.**

6. Singular integrals

(Sections 9.2, 9.3, 9.4 in [GT] Gilbarg -Trudinger, or Chapter 2 in [S] Stein)

For $f \in C_0^\infty(\mathbb{R}^n)$ (or $C_0^\alpha(\mathbb{R}^n)$) we recall that

$$Tf = pv.K * f, \quad K(x) = g(\frac{x}{|x|})|x|^{-n} = g(\omega)r^{-n},$$

with $g : \partial B_1 \to \mathbb{R}$ Lipschitz,

$$\int_{\partial B_1} g\, d\omega = 0.$$

Lemma 6.1.

$$\hat{Tf} = \hat{K}\, \hat{f}$$

for some homogenous of degree 0 *function* $|\hat{K}| \le C(\|g\|_{C^{0,1}}, n)$*.*

Proof. Let $K_\varepsilon := K\chi_{B_{\frac{1}{\varepsilon}} \setminus B_\varepsilon}$. If $\xi = se_1$, $s > 0$, we have

$$\begin{aligned}
\hat{K}_\varepsilon(\xi) &= \int_{B_{\frac{1}{\varepsilon}} \setminus B_\varepsilon} K(x)e^{-ix\,\xi}dx = \int_{B_{\frac{1}{\varepsilon}} \setminus B_\varepsilon} K(x)e^{-isx_1}dx \\
&= \int_{B_{\frac{s}{\varepsilon}} \setminus B_{s\varepsilon}} K(x)e^{-ix_1}dx \\
&= \int_{B_{\frac{s}{\varepsilon}} \setminus B_1} K(x)e^{-ix_1}dx + \int_{B_1 \setminus B_{s\varepsilon}} K(x)(e^{-ix_1} - 1)dx \\
&= I_1(\frac{s}{\varepsilon}) + I_2(s\varepsilon)
\end{aligned}$$

Then $I_2(r)$ converges as $r \to 0$ as

$$|I_2| \le \int_0^1 \|g\|_{L^\infty} r^{-n} C r^2 r^{n-1} dr,$$

while

$$I_1(R) = \frac{1}{i}\int_{B_R \setminus B_1} K_{x_1} e^{-ix_1}dx + \frac{1}{-i}\int_{\partial(B_R \setminus B_1)} K e^{-ix_1}\nu_1 d\sigma,$$

which is integrable as $R \to \infty$. In conclusion

$$|\hat{K}_\varepsilon| \le C, \quad \hat{K}_\varepsilon(\xi) \to \hat{K}(\frac{\xi}{|\xi|}) \quad \text{as } \varepsilon \to 0,$$

hence $\hat{K}_\varepsilon \hat{f} \to \hat{K}\hat{f}$ as $\varepsilon \to 0$.

□

Exercise: Show that

$$\hat{K}(\xi) = \int_{\partial B_1} \left[\frac{\pi i}{2} sign(\xi \cdot \omega) + \log\frac{1}{|\xi \cdot \omega|}\right] g(\omega)\, d\omega.$$

Show that we only need $g \in C^\alpha$ for some $\alpha > 0$.

Corollary: T is a bounded linear operator from L^2 to L^2, i.e.

$$\|Tf\|_{L^2} \le C(K)\|f\|_{L^2}.$$

Theorem 6.2. *T is of weak type $(1,1)$, i.e.*

$$|\{|Tf| \geq \lambda\}| \leq C \frac{\|f\|_{L^1}}{\lambda}.$$

Proof. Reduce the the case $\|f\|_{L^1} = 1$, $\lambda = 1$.

The part where $f < 1$ is good by the L^2 estimate.

How about when $f = \delta_0$? Then we compare f to $\frac{1}{|B_1|}\chi_{B_1}$ and the error is in L^1 outside B_2.

Calderon-Zygmund decomposition: divide the space in dyadiac cubes inductively and keep the cubes Q such that for the first time we satisfy $\fint_Q |f| > 1$. Then we obtain a collection of disjoint cubes Q_i with the property that

$$1 \leq \fint_{Q_i} |f| \leq C(n) \quad \Longrightarrow \quad \sum |Q_i| \leq C, \tag{6.1}$$

and $f \leq 1$ outside $\cup Q_i$. We decompose

$$f = g + b, \quad |g| \leq C, \quad \fint_{Q_i} b = 0,$$

and then write $b = \sum b_i$ with b_i supported in Q_i.

We prove the conclusion for T_ε (with the truncated kernels K_ε) and then let $\varepsilon \to 0$. We have

$$\int_{(2Q_i)^c} |T_\varepsilon b_i| dx \leq C|Q_i|. \tag{6.2}$$

Indeed, this is a scale invariant statement, so we may think that Q_i is the unit cube centered at the origin and we have

$$T_{\tilde{\varepsilon}} b_1(x) = \int_{Q_1} (K_{\tilde{\varepsilon}}(x-y) - K_{\tilde{\varepsilon}}(x))\, b_1(y) dy,$$

hence, if $x \notin 2Q_1$ then

$$|T_{\tilde{\varepsilon}} b_1(x)| \leq C|x|^{-n-1},$$

which proves (6.2). We use that

$$T_\varepsilon b = \sum T_\varepsilon b_i$$

and then (6.2) gives

$$\int_{\mathbb{R}^n - \cup(2Q_i)} |T_\varepsilon b|\, dx \leq C,$$

thus (by (6.1)) we have

$$|\{|T_\varepsilon b| \geq 1/2\}| \leq C.$$

On the other hand

$$\|g\|_{L^2} \leq C\|g\|_{L^1} \leq C\|f\|_{L_1} = C,$$

and the L^2 estimate implies $\|T_\varepsilon g\|_{L^2} \leq C$, thus

$$|\{|T_\varepsilon g| \geq 1/2\}| \leq C.$$

The theorem follows since

$$||\{|T_\varepsilon f| \geq 1\}| \leq |\{|T_\varepsilon g| \geq 1/2\}| + |\{|T_\varepsilon b| \geq 1/2\}| \leq C.$$

$\square$

Definition 6.3. We say that an operator T is of *strong type* (p,p) if there exists a constant C such that

$$\|Tf\|_{L^p} \le C\|f\|_{L^p}.$$

We say that T is of *weak type* (p,p) if

$$|\{|Tf| \ge \lambda\}| \le C\frac{\int |f|^p}{\lambda^p}.$$

Clearly if T is of strong type (p,p) then is of weak type (p,p) as well.

Proposition 6.4 (Marcinkiewitz Interpolation)**.** *Let T be an operator that acts on C_0^1 functions. If T is of weak type (p,p), (q,q) with $p<q$, and T is sublinear*

$$|T(f_1+f_2)| \le |Tf_1| + |Tf_2|,$$

then T it is of strong type (r,r) for all $r \in (p,q)$.

Proof. Let $\underline{f}_\lambda$ represent the truncation of f at the $\pm\lambda$ levels:

$$\underline{f}_\lambda(x) = f(x) \quad \text{if } |f| \le \lambda,$$

$$\underline{f}_\lambda(x) = \lambda\frac{f(x)}{|f(x)|} \quad \text{if } |f| > \lambda.$$

Then we write

$$f = \underline{f}_\lambda + \bar{f}_\lambda.$$

We have

$$|\{|Tf| \ge \lambda\}| \le |\{|T\underline{f}_\lambda| \ge \frac{\lambda}{2}\}| + |\{|T\bar{f}_\lambda| \ge \frac{\lambda}{2}\}|,$$

We denote $a(s) := |\{|f| \ge s\}|$ the distribution function for $|f|$. For the first term we use that T is of weak type (q,q) and find

$$|\{|T\underline{f}_\lambda| \ge \frac{\lambda}{2}\}| \le C\frac{\|\underline{f}_\lambda\|_{L^q}}{\lambda^p} = C\int_0^\lambda a(s)s^{q-1}\lambda^{-q}ds.$$

Similarly,

$$|\{|T\bar{f}_\lambda| \ge \frac{\lambda}{2}\}| \le C\int_\lambda^\infty a(s)s^{p-1}\lambda^{-p}ds.$$

In conclusion

$$|\{|Tf| \ge \lambda\}| \le C\int_0^\lambda a(s)s^{q-1}\lambda^{-q}ds + C\int_\lambda^\infty a(s)s^{p-1}\lambda^{-p}ds.$$

Multiply with λ^{r-1} and integrate. The right hand side becomes

$$\int_0^\infty\int_s^\infty a(s)s^{q-1}\lambda^{r-q-1}d\lambda ds + \int_0^\infty\int_0^s a(s)s^{p-1}\lambda^{r-p-1}d\lambda ds$$

$$\le \int_0^\infty a(s)s^{r-1}ds = \|f\|_{L^r}.$$

we obtain the desired conclusion. □

Theorem 6.5 (Calderon-Zygmund)**.** *Let T be a singular integral operator.*

Then T is of strong type (p,p) for all $p \in (1,\infty)$, and of weak type $(1,1)$.

Proof. By Marcinkiewitz Interpolation we know the result for $p \in (1, 2]$.

For $p > 2$ we use a duality argument. Let $p' \in (1, 2)$ be its conjugate. Let φ, $\psi \in C_0^\infty$, and denote by $\bar{K}(x) := K(-x)$. Then, we use the result for p' and Holder inequality:

$$C\|\varphi\|_{L^{p'}}\|\psi\|_{L^p} \geq \|T_{\bar{K}}\varphi\|_{L^{p'}}\|\psi\|_{L^p} \geq \int (T_{\bar{K}}\varphi)\,\psi dx = \int \varphi\,(T_K\psi)dx.$$

Now the conclusion follows by taking the supremum over all φ with $\|\varphi\|_{L^{p'}} = 1$. □

6.1. **Problems.**

1. a) The *Riesz transform* R_j (in the x_j direction) is defined as the singular integral operator with kernel $x_j/|x|^{n+1}$. Show that

$$\hat{R}_j(\xi) = c_n \frac{\xi_j}{|\xi|},$$

with c_n a complex constant with zero real part.

In 1D the Riesz transform (with kernel $1/x$) is known as the *Hilbert transform.*

b) Assume that u and v are defined in the upper half plane and they satisfy the Cauchy Riemann equations

$$u_x = v_y, \quad u_y = -v_x.$$

Assume that u, v and their derivatives decay polynomially at infinity. By taking the Fourier transform in the x variable show that $u(x, 0)$ is the Hilbert transform of $v(x, 0)$.

2. a) If f is supported in B_1 and $|f| \log(2 + |f|)$ is integrable in B_1, then Tf is integrable in B_1 (follow the proof of the Marcinkiewitz interpolation theorem).

b) By inspecting the proof of the Marcinkiewitz interpolation theorem (and the duality argument) show that the constant C_p in the Calderon-Zygmund estimate

$$\|Tf\|_{L^p} \leq C_p\|f\|_{L^p}$$

satisfies

$$C_p \leq C_0 \frac{1}{p-1} \quad \text{as } p \to 1, \text{ and} \quad C_p \leq C_0\, p \quad \text{as } p \to \infty,$$

with C_0 independent of p.

c) If f is bounded in B_1 then $e^{\delta|Tf|}$ is integrable in B_1 for some small $\delta > 0$ depending on the kernel K.

(Expand the exponential as a power series and apply part b.)

3. Let $\alpha \in (0, n)$ and define the Riesz potential of order α of f as

$$I_\alpha(f) := f * |x|^{\alpha - n}.$$

(when $\alpha = 2$ this is the same as the Newtonian potential).

a) Show that I is of weak type (p, q) with $\frac{1}{q} := \frac{1}{p} - \frac{\alpha}{n}$ with $p \in [1, \frac{n}{\alpha})$ that is

$$|\{|I_\alpha f| > \lambda\}| \leq C\left(\frac{\|f\|_{L^p}}{\lambda}\right)^q.$$

(Hint: reduce to the case $\|f\|_{L^p} = 1$, $\lambda = 1$ after scaling and then decompose the kernel $|x|^{\alpha-n}$ into the part supported in B_1 and the part supported in B_1^c.)

b) It turns out by a variant of Markinciewitz interpolation theorem that I_α is of strong type (p, q) except for the end-points $p = 1$, $p = \frac{n}{\alpha}$ (do not prove this fact).

Show that this implies the Sobolev inequality for $p \in (1, n)$ by bounding $|f|$ by $CI_1(|\nabla f|)$ for f with compact support.

7. $W^{2,p}$ ESTIMATES

(Sections 9.5 and 9.6 from Gilbarg-Trudinger)

In this section we assume

$$p \in (1, \infty).$$

Theorem 7.1. *If $\triangle u = f$ in B_1 with $f \in L^p$ then*

$$\|u\|_{W^{2,p}(B_{1/2})} \leq C(\|u\|_{L^p(B_1)} + \|f\|_{L^p(B_1)}),$$

with C depending on n and p.

Proof. Indeed, extend $f = 0$ outside B_1, and then

$$u = -f * \Gamma + h \qquad \text{with} \quad \triangle h = 0 \qquad \text{in } B_1.$$

Since $\Gamma \in L^1(B_2)$ we find $\|f * \Gamma\|_{L^p(B_1)} \leq C\|f\|_{L^p(B_1)}$ hence

$$\|h\|_{L^p(B_1)} \leq C(\|u\|_{L^p} + \|f\|_{L^p}).$$

Since h is harmonic in B_1 we have

$$\|h\|_{C^2(B_{1/2})} \leq C\|h\|_{L^p(B_1)},$$

and now the result follows from the estimates of the previous section applied to the second derivatives of $f * \Gamma$ (which are singular integrals) .

□

Next we extend the result of Theorem 7.1 to operators with variable coefficients in a similar way we did for the Schauder $C^{2,\alpha}$ estimates. We provide some of the details.

We consider operators

$$Lu = a^{ij}u_{ij} + b^i u_i + cu, \tag{7.1}$$

with a^{ij} uniformly elliptic, continuous and b^i, c bounded.

Theorem 7.2 ($W^{2,p}$ estimates)**.** *If $u \in W^{2,p}$ satisfies*

$$Lu = f \quad \text{ in } B_1,$$

then

$$\|u\|_{W^{2,p}(B_{1/2})} \leq C(\|u\|_{L^p(B_1)} + \|f\|_{L^p(B_1)}),$$

with C depending on the coefficients and the ellipticity.

Theorem 7.2 follows by glueing together the $W^{2,p}$ estimates in small balls where the coefficients have small oscillation. Thus, after a translation, dilation, and affine transformation (whose norm depends on the ellipticity constants) it suffices to show the estimates when L is a perturbation of $\triangle$.

Lemma 7.3. *If $|a_{ij} - \delta_i^j|$, $|b^i|$, $|c| \leq \delta$, then the conclusion holds for a universal constant C.*

Proof. Step 1: We show that

$$\|D^2u\|_{L^p(B_{1/2})} \leq C(\|u\|_{L^p(B_1)} + \|f\|_{L^p(B_1)}) + \mu \max_{B_{\frac14}(x_0) \subset B_{\frac34}} \|D^2u\|_{L^p(B_{1/4}(x_0))},$$

with $\mu = 2^{-3}$. We write the equation as $\triangle u = f + \mathit{error}$ and find

$$\|D^2 u\|_{L^p(B_{1/2})} \leq C(\|u\|_{L^p(B_1)} + \|f\|_{L^p(B_1)} + \delta \|D^2 u\|_{L^p(B_1)} + \delta \|\nabla u\|_{L^p(B_1)}).$$

We use the interpolation inequality (exercise after Sobolev spaces)

$$\|\nabla u\|_{L^p(B_1)} \leq \|D^2 u\|_{L^p(B_1)} + C_p \|u\|_{L^p(B_1)},$$

and get the conclusion, provided that δ is chosen small.

Rescaled conclusion:

$$\|D^2 u\|_{L^p(B_{r/2})} \leq C(r^{-2}\|u\|_{L^p(B_r)} + \|f\|_{L^p(B_r)}) + \mu \max \|D^2 u\|_{L^p(B_{r/4}(x_0))}.$$

As before we can construct a sequence of balls $B_k := B_{2^{-k}}(x_k) \subset B_1$ so that

$$a_k \leq C_1 2^{2k} + \mu\, a_{k+1}, \qquad a_k := \|D^2 u\|_{L^p(B_k)}.$$

We conclude that $a_1 \leq C(C_1)$, otherwise $a_k \to \infty$ and we contradict the assumption that $u \in W^{2,p}(B_1)$.

□

Theorem 7.4 (Boundary version). *If $u \in W^{2,p}(B_1^+) \cap W_0^{1,p}(B_1^+)$ satisfies*

$$Lu = f \quad \text{in } B_1^+,$$

then

$$\|u\|_{W^{2,p}(B_{1/2}^+)} \leq C(\|u\|_{L^p(B_1^+)} + \|f\|_{L^p(B_1^+)}),$$

with C depending on the coefficients and the ellipticity.

By $u \in W_0^{1,p}(B_1^+)$ we understand that u vanishes on $\{x_n = 0\}$ in the sense of traces of $W^{1,p}$ functions.

The proof of Theorem 7.4 is identical to that of Theorem 7.2 (and Lemma 7.3), after we establish the boundary version of Theorem 7.1 for the Laplace equation (see Problem 1 below.)

If the boundary of a domain Ω is of class $C^{1,1}$, than it can be made flat locally by a $C^{1,1}$ domain deformation which keeps the operator L in the same class (see Problem 2 below). We obtain the following global version of the theorem above.

Theorem 7.5 (Global version). *If $u, \varphi \in W^{2,p}(\Omega)$, $u - \varphi \in W_0^{1,p}(\Omega)$ with $\partial\Omega \in C^{1,1}$ satisfies*

$$Lu = f \quad \text{in } \Omega,$$

then

$$\|u\|_{W^{2,p}(\Omega)} \leq C(\|u\|_{L^p(\Omega)} + \|f\|_{L^p(\Omega)} + \|\varphi\|_{W^{2,p}(\Omega)}),$$

with C depending on the coefficients and the ellipticity.

7.1. **Existence and Uniqueness.** Assume that $c \leq c(\lambda, \Lambda, n)$ sufficiently small so that the maximum principle works.

Theorem 7.6 (Existence)**.** *Let $\varphi \in W^{2,p}(\Omega)$, $f \in L^p(\Omega)$, and $\partial\Omega \in C^{1,1}$. There exists a unique solution $u \in W^{2,p}(\Omega)$ to*
$$Lu = f \quad \text{in } \Omega, \qquad u - \varphi \in W_0^{1,p}(\Omega),$$
and
$$\|u\|_{W^{2,p}(\Omega)} \leq C(\|f\|_{L^p(\Omega)} + \|\varphi\|_{W^{2,p}(\Omega)}),$$
with C depending on the coefficients and the ellipticity.

One difference with the $C^{2,\alpha}$ estimates is that uniqueness does not follow right away from the maximum principle. The reason is that at a maximum point (or contact point with a barrier) we cannot talk about the values of the second derivatives since they are defined only as L^p functions. It turn out that if $p \geq n$ then, a variant of the maximum principle holds via the Alexandrov-Bakelman-Pucci theorem. This is known as the "ABP estimate" (see Theorem 9.1 in [GT]) which we will discuss later on.

In order to prove the uniqueness for all $p > 1$ we need the following higher regularity lemma:

Lemma 7.7. *If $u \in W^{2,p}_{loc}(B_1)$ satisfies $Lu = f$ with $f \in L^q$ for some $q > p$ then $u \in W^{2,q}_{loc}$.*

Proof. After a dilation and an affine transformation we may assume that $\|a^{ij} - \delta^i_j\|_{L^\infty}$ is sufficiently small. Let $\eta \in C_0^\infty$ be a cutoff function, and then $v = \eta u$ solves an equation of the type (by Sobolev embedding)
$$a^{ij} v_{ij} = g \in L^{p+\delta}, \quad v \in W^{2,p}.$$
This can be written as
$$\triangle v = g + (\delta^i_j - a^{ij}) v_{ij},$$
and since v has compact support we can rewrite it as
$$v = T_1 g + T_2 v$$
with $T_1 g \in W^{2,p+\delta}$ and $T_2 : W^{2,p} \to W^{2,p}$ with $\|T_2\| \leq 1/2$. Moreover T_2 is a contraction also on the $W^{2,p+\delta}$ and the contraction principle implies the desired conclusion.

$\square$

A similar statement for the lemma above holds up to the boundary if we assume $u \in W_0^{1,p}(\Omega)$.

Now we provide the proof of Theorem 7.6.

Uniqueness: It suffices to assume $f = \varphi = 0$, and then by the Lemma above $u \in W^{2,q}(\Omega)$ for all $q > p$. Now we can apply the ABP maximum principle (we need $q \geq n$).

Existence: We can obtain the solution by approximating the data and coefficients by smooth data and then by taking limits of the corresponding solutions which exist by the Schauder theory. In order to extract a convergent subsequence (to the solution of our problem) we need a uniform bound for $\|u\|_{W^{2,p}}$. By the global $W^{2,p}$

estimates we only need to show $\|u\|_{L^p}$ is bounded in terms of $\|f\|_{L^p}$, $\|\varphi\|_{W^{2,p}}$. This follows by compactness. Otherwise we can find a solution to $Lu = 0$, $\varphi = 0$ with $\|u\|_{L^p} = 1$ which contradicts uniqueness.

7.2. **Problems.**

1. If $\triangle u = f$ in B_1^+ with $f \in L^1$, and u vanishes on $\{x_n = 0\} \cap B_1$ in the sense of $W^{1,1}$ traces (i.e. $u \in W_0^{1,1}(B_1^+)$) then the odd extension $\tilde{u}$ of u with respect to the $\{x_n = 0\}$ satisfies

$$\triangle \tilde{u} = \tilde{f} \quad \text{in} \quad B_1,$$

where $\tilde{f}$ is the odd reflection of f.

Conclude the boundary version of Theorem 7.1:

If $\triangle u = f$ in B_1^+ with $f \in L^p$, and $u \in W_0^{1,p}(B_1^+)$ then

$$\|u\|_{W^{2,p}(B^+_{1/2})} \leq C(\|u\|_{L^p(B_1^+)} + \|f\|_{L^p(B_1^+)}),$$

with C depending on n and p (here $p \in (1, \infty)$).

2. a) Show that if $\Phi \in C^{1,1}$ is a diffeomorphism and $u \in W^{2,p}$ solves $Lu = f \in L^p$ with L as in (7.1), then $v(x) = u(\Phi(x))$ solves $\tilde{L}v = \tilde{f} \in L^p$, and $\tilde{L}$ satisfies the conditions in (7.1) as well.

b) Show that the L^∞ bounds on the lower order coefficients $b(x)$, $c(x)$ in (7.1) can be relaxed by Sobolev imbedding to $b \in L^q$ with $q > n$ if $p \leq n$ or $q = p$ if $p > n$, and $c \in L^r$ with $r > n/2$ if $p \leq n/2$ or $r = p$ if $p > n/2$.

c) Deduce that the global estimate holds for the $W^{2,p}$ estimate holds for domains Ω with $\partial\Omega \in C^{1,\alpha}$ with $\alpha > 1 - \frac{1}{p}$.

8. Sobolev spaces

(Chapter 7 in Gilbarg-Trudinger, or Chapter 5 in Evans)

Definition 8.1 (Weak derivatives)**.** Let $\Omega \subset \mathbb{R}^n$ be an open set. Assume that $u, v \in L^1_{loc}(\Omega)$ (or more generally locally bounded measures). We say that $u_i = v$ in Ω if

$$\int u\, \varphi_i = -\int v\, \varphi, \qquad \forall\, \varphi \in C_0^\infty(\Omega).$$

Similarly we can define higher order weak derivatives.

Examples:

1) In 1D if $u(x) = |x|^\alpha$ with $\alpha \in (0,1)$ then $u'(x) = \alpha\ sign(x)\, |x|^{\alpha-1}$. If $\alpha < 0$ then we cannot talk about u' as a function (or measure) in an interval that contains 0.

In higher dimensions we can consider first derivatives of $|x|^\alpha$ in domains containing the origin only if $\alpha > 1 - n$.

2) In 1D if $u(x) = \chi_{[0,\infty)}$ then u' is a measure supported at the origin, i.e $u' = \delta_0$.

More generally, if $u = \chi_E$ for a set $E \subset \mathbb{R}^n$ with smooth boundary then u_i is a measure supported on ∂E. Precisely, divergence theorem shows that $u_i = -\nu^i d\sigma$ where ν^i is the ith component of the outer unit normal ν to ∂E, and $d\sigma$ is the surface measure of the boundary of E.

3) If u is a Lipschitz function in an interval then $u^h(x) := \frac{1}{h}(u(x+h) - u(x))$ is a bounded function in x. By compactness, we can find a sequence of $h_n \to 0$ such that u^{h_n} converge weakly to an L^∞ function. This limiting function must be the weak derivative of u. Thus, any Lipschitz function has a derivative which is an L^∞ function.

4) More generally, a function posses a weak derivative in the x_i direction in a domain Ω if the discrete differences

$$D_i^h u(x) := \frac{1}{h}(u(x + he_i) - u(x))$$

are uniformly integrable (as $h \to 0$) on compact sets of Ω.

Important properties:

1) Let ρ_ε be a mollifier supported in B_ε. Then

$$(u * \rho_\varepsilon)_i = u_i * \rho_\varepsilon. \tag{8.1}$$

2) Stability under weak limits: If u_k is a sequence such that

$$u_k \rightharpoonup u, \qquad \partial_{x_i} u_k \rightharpoonup v \quad \Longrightarrow \quad \partial_{x_i} u = v. \tag{8.2}$$

We recall that $u_k \rightharpoonup u$ in Ω by definition means that

$$\int u_k\, \varphi\, dx \to \int u\, \varphi\, dx \quad \forall \varphi \in C_0(\Omega).$$

Consequences of 1), 2) above by working with $u * \rho_\varepsilon$ and then letting $\varepsilon \to 0$:

1)
$$(u\xi)_i = u_i \xi + u \xi_i, \qquad \xi \in C^1,$$
2)
$$(u \circ \Psi)_i = (u_j \circ \Psi)\Psi_i^j, \qquad \Psi \text{ a } C^1 \text{ diffeomorphism},$$
3)
$$(\varphi(u))_i = \varphi'(u) u_i, \qquad \varphi \in C^1, \quad |\varphi'| \le C.$$

The formulas above can hold in more general situations, for example whenever the right hand sides are integrable (we could take $\xi \in C^{0,1}$ or $\Phi \in C^{0,1}$). An important case is when in 3) we take $\varphi(t) = t^+$ which gives

$$(u^+)_i = \chi_{\{u>0\}}\, u_i. \tag{8.3}$$

This can be seen by approximating t^+ by a sequence of C^1 functions $\varphi_\varepsilon(t)$ which are 0 in $(-\infty, \varepsilon]$ and equal to t on $[2\varepsilon, \infty)$.

Definition 8.2 (Sobolev spaces)**.**
1) For $1 \le p \le \infty$ and $\Omega \subset \mathbb{R}^n$ we define $W^{1,p}(\Omega)$ as
$$W^{1,p}(\Omega) := \{u : \Omega \to \mathbb{R} | \quad u,\, u_i \in L^p(\Omega)\},$$
$$\|u\|_{W^{1,p}} := \|u\|_{L^p} + \sum_{i=1}^n \|u_i\|_{L^p}.$$

2) The space of functions of bounded variations $BV(\Omega)$ is similar to $W^{1,1}$ except that the derivatives are required to be only bounded measures (instead of L^1 functions)
$$BV(\Omega) := \big\{u : \Omega \to \mathbb{R} | \quad u \in L^1, \quad u_i \quad \text{bounded measures}\big\} \supset W^{1,1}(\Omega),$$
$$\|u\|_{BV} := \|u\|_{L^1} + \int |Du|, \qquad \int |Du| := \sup_{g \in C_0^\infty(\Omega), |g| \le 1} \int u \, div\, g.$$

3)
$$W_0^{1,p}(\Omega) \quad \text{is defined as the completion of } C_0^\infty \text{ into } W^{1,p}(\Omega).$$

Similarly one can define the spaces $W^{k,p}(\Omega)$ (and $W_0^{k,p}(\Omega)$) involving higher order derivatives up to the kth order, which are required to be L^p functions.

Proposition 8.3. *$W^{1,p}$, $W_0^{1,p}$, BV are Banach spaces.*

The proof follows easily from the stability under limits of weak derivatives (8.2).

Notation. The space $W^{1,2}(\Omega)$ is a Hilbert space and it is often denoted by
$$H^1(\Omega) := W^{1,2}(\Omega), \qquad H_0^1(\Omega) := W_0^{1,2}(\Omega).$$
The inner product on H^1 is
$$u \cdot v = \int_\Omega uv + \nabla u \cdot \nabla v.$$
Similarly $H^k := W^{k,2}$.

Approximation. 1) If $u \in W^{1,p}(\Omega)$ then by (8.1)
$$u * \rho_\varepsilon \to u \qquad \text{in } W^{1,p}(\Omega') \text{ as } \varepsilon \to 0, \text{ with} \qquad \Omega' \subset\subset \Omega.$$

Notice that if u has compact support then we can take ε sufficiently small such that $u * \rho_\varepsilon$ is as close as we want to u in the $W^{1,p}$ distance, and its support is included in a given open set containing *supp* u.

2) Next we show that there exists a sequence $\varphi_k \in C^\infty(\Omega)$ such that
$$\varphi_k \to u \quad \text{in } W^{1,p}(\Omega).$$
Indeed, we can use a partition of unity and write
$$u = \sum_1^\infty u^m, \qquad u^m \in W^{1,p}$$
with *supp* $u^m \subset\subset \Omega$ and having the finite overlapping property. By 1) we can find φ^m such that
$$\|u^m - \varphi^m\|_{W^{1,p}} \le \varepsilon 2^{-m}.$$
Then we define $\varphi := \sum_m \varphi^m \in C^\infty$, and the inequlity above gives the desired inequality $\|u - \varphi\|_{W^{1,p}} \le \varepsilon$.

3) We remark that when $\Omega = \mathbb{R}^n$ then the spaces $W^{k,p}$ and $W_0^{k,p}$ coincide (see Problem 1).

Extension. When Ω is a Lipschitz domain (its boundary is given locally by a Lipschitz graph) then we can extend a function $u \in W^{1,p}(\Omega)$ in a neighborhood of Ω without increasing too much its $W^{1,p}$ norm.

Proposition 8.4. *Assume that Ω is a bounded Lipschitz domain and $u \in W^{1,p}(\Omega)$. There exists a function $\tilde u \in W_0^{1,p}(\tilde\Omega)$ which coincides with u in $\Omega \subset \tilde\Omega$ such that*
$$\|\tilde u\|_{W^{1,p}} \le C(\Omega)\, \|u\|_{W^{1,p}(\Omega)}.$$

As a consequence, when Ω is Lipschitz we can define the convolution $u * \rho_\varepsilon$ also near $\partial\Omega$, by first extending u. In particular we can approximate $u \in W^{1,p}(\Omega)$ with functions which are in $C^\infty(\bar\Omega)$ (smooth up to the boundary of Ω).

Sketch of proof. Locally, after a bi-Lipschitz deformation we can reduce to the case when $\partial\Omega$ is flat, say $\Omega \cap B_1 = \{x_n > 0\}$. Then we can reflect u evenly in the x_n variable. Going back to the original picture, after patching the regions around $\partial\Omega$ with a partition of unity, we obtain a function $v \in W^{1,p}(\tilde\Omega)$ with $\Omega \subset\subset \tilde\Omega$ and
$$\|v\|_{W^{1,p}(\tilde\Omega)} \le C(\Omega)\, \|u\|_{W^{1,p}(\Omega)}.$$
Now we take $\tilde u = \eta v$ with η a cutoff function which is 1 on Ω.

$\square$

Trace. When $u \in W^{1,p}(\Omega)$, and Ω is a Lipschitz domain, we can define the boundary values of u on $\partial\Omega$ (the trace of u). After an bi-Lipschitz deformation we can assume that locally Ω is the cylinder $\Omega = B_1' \times [0,1]$. When $u \in C^1(\bar\Omega)$ we have
$$|u(x', y_n) - u(x', z_n)| \le \int_0^t |u_n(x', x_n)| dx_n, \qquad \forall y_n, z_n \in [0,t]$$
$$\int_{B_1'} |u(x', y_n) - u(x', z_n)|^p \, dx' \le \int_{B_1' \times [0,t]} |u_n|^p dx.$$

By averaging over the variables $y_n \in [0,t]$ and $z_n \in [0,s]$ with $s \le t$ we find

$$\int_{B_1'} \left| \fint_0^t u(x',x_n)dx_n - \fint_0^s u(x',x_n)dx_n \right|^p dx' \le \int_{B_1' \times [0,t]} |u_n|^p dx.$$

By approximation this inequality is valid for any $u \in W^{1,p}(\Omega)$ and we obtain that the limit

$$\lim_{t \to 0} \fint_0^t u(x',x_n)dx_n =: u(x',0)$$

exists in the sense of L^p functions. The trace of u is linear in u and it coincides with the boundary values whenever u is continuous up to the boundary.

We remark that $u \in W_0^{1,p}(\Omega)$ if and only if its trace vanishes on $\partial\Omega$.

8.1. **Sobolev embedding.**

Theorem 8.5. *a) (Sobolev inequality)*
If $1 \le p < n$ and $u \in W_0^{1,p}(\mathbb{R}^n)$ then

$$C_{n,p} \left(\int |\nabla u|^p dx \right)^{\frac{1}{p}} \ge \left(\int u^{p^*} dx \right)^{\frac{1}{p^*}}, \qquad \frac{1}{p^*} := \frac{1}{p} - \frac{1}{n}.$$

b) (Morrey inequality)
If $p > n$ and $u \in W^{1,p}(B_1)$ then

$$C_{n,p} \|\nabla u\|_{L^p(B_1)} \ge [u]_{C^\alpha(B_1)}, \qquad \alpha := 1 - \frac{n}{p}.$$

In particular this Theorem implies that the following embeddings are continuous $W^{1,p}(B_1) \to L^{p^*}(B_1)$ if $p < n$ and $W^{1,p}(B_1) \to C^{1-\frac{n}{p}}(B_1)$ if $p > n$.

In the borderline case $p = n$, $W^{1,n}$ embeds into the BMO space (see Problem 7).

Proof of a). It suffices to consider the case $p = 1$. The general follows from $p = 1$ by replacing u by $u^{\frac{n-1}{n}p^*}$.

We may assume that $u \in C_0^\infty$ as the general statement follows by approximation.

Claim: When $p = 1$ Sobolev inequality is equivalent to the isoperimetric inequality (when $u = \chi_E$, and $\partial E \in C^1$)

$$C \; \mathcal{H}^{n-1}(\partial E) \ge |E|^{\frac{n-1}{n}}.$$

For the converse we write the coarea formula (say $u \ge 0$)

$$\int |\nabla u| dx = \int_0^\infty \mathcal{H}^{n-1}(\partial\{u > t\}) dt \; \ge \; c \int_0^\infty a(t)^{\frac{n-1}{n}} dt,$$

with $a(t) := |\{u > t\}|$ a decreasing function of t. We use the inequality

$$\int_0^\infty a(t)^{\frac{n-1}{n}} dt \ge \left(\int_0^\infty a(t) \; t^{\frac{1}{n-1}} dt \right)^{\frac{n-1}{n}}, \tag{8.4}$$

and obtain the desired conclusion

$$\int |\nabla u| dx \ge c \|u\|_{L^{\frac{n}{n-1}}}.$$

To check inequality (8.4), we may assume that the left hand side is 1, and since a is decreasing we have

$$t \, a(t)^{\frac{n-1}{n}} \le 1 \qquad \Longrightarrow \qquad a(t)^{\frac{n-1}{n}} \ge a(t) \, t^{\frac{1}{n-1}}.$$

Proof of the isoperimetric ineqaulity. We want to show that if $|E| = 1$ then the projection of ∂E (which is the same of projection of E) onto some hyperplane is greater than some c_n small. In 2D it is not difficult to see that the product of the projections onto the coordinate axes is greater than 1. We want to use the same argument in $n \geq 3$.

Key inequality: Assume f_i, $i = 1, .., n$ are functions with each f_i independent of the x_i variable. Then

$$\int \Pi_{i=1}^n \ f_i \, dx \leq \ \Pi_{i=1}^n \ \|f_i\|_{L^{n-1}(\mathbb{R}^{n-1})},$$

where $\|f_i\|_{L^{n-1}(\mathbb{R}^{n-1})}$ denotes the L^{n-1} norm in the depending $n-1$ variables (except x_i).

The proof follows by induction.

We obtain the desired conclusion by choosing f_i to be the projections of χ_E along the x_i variable.

Direct proof of Sobolev inequality. Denote by $f_i^{n-1} := \int_{\mathbb{R}} |u_i| dx_i$ which is independent of the x_i variable, and clearly

$$|u|^{\frac{1}{n-1}} \leq f_i, \qquad \|f_i\|_{L^{n-1}(\mathbb{R}^{n-1})} \leq \|\nabla u\|_{L^{n-1}(\mathbb{R}^n)}.$$

Alternate proof: We use Calderon-Zygmund cube decomposition and decompose E (with $|E| = 1$) into a countable union of disjoint cubes Q_i such that

$$c_n \leq \frac{|Q_i \cap E|}{|Q_i|} \leq 1 - c_n.$$

It suffices to show that

$$\mathcal{H}^{n-1}(\partial E \cap Q_i) \geq c_0 |Q_i|^{\frac{n-1}{n}},$$

and then the result follows by summing over i.

To prove the last inequality, after scaling, we may assume that Q_i is the unit cube and that both E and its complement E^c cover at least a fixed proportion of the cube. Now there are several ways to proceed, for example one can use Poincare inequality (see below) or find directly a unit direction along which the projections of E and E^c overlap on a set of positive measure.

$\square$

Theorem 8.6 (Poincare inequality)**.** *Let $u \in W^{1,1}(B_1)$. Then*

$$\int_{B_1 \times B_1} |u(x) - u(y)| \ dxdy \leq C \int_{B_1} |\nabla u| dx.$$

In particular

$$\int_{B_1} \left| u(x) - \fint_{B_1} u \right| dx \leq C_n \int_{B_1} |\nabla u| dx.$$

Proof. We prove the inequality for smooth functions and then the result follows by approximation. Pick $y \in B_1$ and then writing $x = y + r\omega$ with $\omega \in \partial B_1$ we find

$$u(x) - u(y) = u(y + r\omega) - u(y) = \int_0^r \nabla u(y + t\omega) \cdot \omega \, dt,$$

or

$$|u(x) - u(y)| \leq \int_0^r |\nabla u(y + t\omega)|\, dt \tag{8.5}$$

We integrate in $x \in B_1$ using polar coordinates: for each ω we multiply the inequality with r^{n-1} and then integrate in r from 0 till the value r_ω where the ray $y + r\omega$ hits ∂B_1. We find

$$\int_{B_1} |u(x) - u(y)| dx \leq C(n) \int_{\partial B_1} \int_0^{r_\omega} |\nabla u(y + t\omega)|\; dt d\omega$$

or

$$\int_{B_1} |u(x) - u(y)| dx \leq C(n) \int_{B_1} \frac{|\nabla u(x)|}{|x - y|^{n-1}} dx. \tag{8.6}$$

We integrate in y (the weight $|x - y|^{1-n}$ is in L^1) and obtain

$$\int_{B_1 \times B_1} |u(x) - u(y)|\, dxdy \leq C'(n) \int_{B_1} |\nabla u| dx.$$

If we divide by $|B_1|$, then the integration in y is an average over y and, by triangle inequality, the left hand side is greater

$$\int_{B_1} \left| u(x) - \fint_{B_1} u \right| dx,$$

which gives the desired inequality.

□

Remark 8.7. a) From (8.5) we see that

$$|u(x) - u(y)|^p \leq C(p) \int_0^r |\nabla u(y + t\omega)|^p\, dt,$$

Thus the Poincare inequalities hold if $u \in W^{1,p}$, $p \geq 1$, and all the absolute values are raised to power p.

The inequalities hold if we replace B_1 by a convex set $D \subset B_1$ as long as we adjust the constants C to depend on $|D|$ as well. (Or more generally for Lipschitz domains D.)

Proof of Theorem 8.5 b). The rescaling of Poincare inequality says that

$$\fint_{B_r} \left| u(x) - \fint_{B_r} u \right| dx \leq Cr \fint_{B_r} |\nabla u| dx$$

and by Holder inequality

$$\fint_{B_r} \left| u(x) - \fint_{B_r} u \right| dx \leq Cr^{1-n} r^{n(1-\frac{1}{p})} \left(\int_{B_r} |\nabla u|^p dx \right)^{\frac{1}{p}},$$

thus, for any balls $B_{r/2}(x_1) \subset B_r(x_0)$ we have

$$\left| \fint_{B_r(x_0)} u - \fint_{B_{r/2}(x_1)} u \right| dx \leq Cr^{1-\frac{n}{p}} \|\nabla u\|_{L^p}.$$

The averages above are taken only in the regions where the balls intersect B_1. From here we conclude that

$$[u]_{C^\alpha(B_1)} \le C\|\nabla u\|_{L^p(B_1)}.$$

□

8.2. **Compactness.**

Theorem 8.8. *Let $p < n$ and let $u_k \in W^{1,p}(B_1)$ be a sequence of functions with bounded $W^{1,p}$ norm. Then there exists a convergent subsequence in any $L^q(B_1)$ with $q < p^*$.*

Proof. After extending u_k's outside B_1 we may assume that they are uniformly bounded in $W_0^{1,p}(B_2)$. If $u \in W_0^{1,p}(B_2)$ then

$$\|u - u * \rho_\varepsilon\|_{L^p} \le \sup_{|\tau|\le\varepsilon} \|u(\cdot) - u(\cdot+\tau)\|_{L^p} \le C\varepsilon \,\|\nabla u\|_{L^p},$$

thus u is well approximated in L^p by $u * \rho_\varepsilon$ (with error $C\varepsilon\|u\|_{W^{1,p}}$).

On the other hand, for fixed ε, these functions are uniformly Lipschitz since

$$\|\nabla(u * \rho_\varepsilon)\|_{L^\infty} \le \|u * \nabla\rho_\varepsilon\|_{L^\infty} \le \|u\|_{L^1}\|D\rho_\varepsilon\|_{L^\infty} \le C\varepsilon^{-(n+1)}\|u\|_{L^p}.$$

Now the compactness in L^p follows by Arzela-Ascoli theorem.

By Holder inequality, a sequence that converges in L^p and has uniformly bounded norm in L^{p^*}, must converge in L^q for all $q \in [p, p^*)$.

□

Discrete differences.

The following theorem gives a criterion for an L^p function to belong to $W^{1,p}$.

Theorem 8.9. *Let $p \in (1, \infty]$. Then*

$$u_i \in L^p_{loc}(\Omega) \quad \Longleftrightarrow D_i^h u(x) := \frac{1}{h}(u(x+he_i)-u(x)) \quad \textit{is uniformly bounded in } L^p_{loc}(\Omega),$$

for all small h.

Proof. Indeed, for $\Omega' \subset\subset \Omega$ and all small h we find (by integrating (8.5))

$$\|D_i^h u\|_{L^p(\Omega')} \le \|u_i\|_{L^p(\Omega)}.$$

Conversely, since $D_i^h u \rightharpoonup u_i$ we find

$$\|u_i\|_{L^p(\Omega')} \le \liminf \|D_i^h u\|_{L^p(\Omega')}.$$

□

The argument above shows that

$$W^{1,\infty}(\Omega) = C^{0,1}(\Omega).$$

8.3. **Problems.**

1. Show that if $p < \infty$, then
$$W^{1,p}(\mathbb{R}^n) = W_0^{1,p}(\mathbb{R}^n).$$

2. Use compactness to show that for any $\delta > 0$ there exists $C = C(n, p, \delta)$ such that
$$\|\nabla u\|_{L^p(B_1)} \leq \delta \|D^2 u\|_{L^p(B_1)} + C\|u\|_{L^p(B_1)}.$$

3. a) Let Ω be a Lipschitz domain and $u \in W^{1,1}(\Omega)$. Deduce a version of Morrey's inequality at a boundary point: for a point $x_0 \in \partial\Omega$ we have
$$\fint_{B_r(x_0)\cap\Omega} |u - a| dx \leq C r^{1-n} \int_{B_{Cr}(x_0)\cap\Omega} |\nabla u| dx, \quad \text{for some } a \in \mathbb{R}.$$

b) By using a Vitalli covering, show that for $\mathcal{H}^{n-1}$ a.e. $x_0 \in \partial\Omega$ we have
$$\lim_{r\to 0} r^{1-n} \int_{B_r(x_0)\cap\Omega} |\nabla u| dx = 0.$$

c) Define the trace of u on $\partial\Omega$ pointwise as
$$u(x_0) := \lim_{r\to 0} \fint_{B_r(x_0)\cap\Omega} u,$$
and conclude that for $\mathcal{H}^{n-1}$ a.e. $x_0 \in \partial\Omega$ we have
$$\lim_{r\to 0} \fint_{B_r(x_0)\cap\Omega} |u(x) - u(x_0)| = 0.$$

4. Assume that u has average 0 in B_1 and that the Poincare inequality
$$\|u\|_{L^q(B_1)} \leq C_{p,q} \|\nabla u\|_{L^p(B_1)}.$$
holds for some $q = p \in [1, n)$. Extend u to a $W_0^{1,p}(B_2)$ function and then apply Sobolev inequality to deduce that Poincare inequality holds also with $q = p^*$ and therefore for all $q \in [1, p^*]$. Conclude that all the inequalities with $q \in [1, p^*]$, $p \in [1, n)$, follow from the case $p = q = 1$.

5. a) Show that if $u \in W^{1,p}_{loc}$ with $p > n$ then u is differentiable at any Lebesgue point of ∇u.

b) Show that if $u \in W^{2,p}_{loc}$ with $p > n/2$ then u is twice differentiable a.e. (meaning that $u = P_{x_0} + o(|x - x_0|^2)$ for a.e. x_0, P_{x_0} quadratic polynomial).

6. a) Show that if u is convex in B_1 then u_{ij} are bounded measures locally, i.e. $\nabla u \in BV$.

b) Show that u is twice differentiable a.e. (known as Alexandrov Theorem).

7. The space BMO consists of $L^1_{loc}(\mathbb{R}^n)$ functions f such that

$$\fint_{B_r(x)} \left| f - \fint_{B_r(x)} f \right| dx \le M, \quad \forall B_r(x),$$

for a constant $M > 0$ (the smallest such M is the BMO seminorm of f.)

a) Show that we may replace balls with cubes in the definition above after changing M by a factor.

b) Assume that $\fint_{Q_1} f = 0$, $M = 1$, and using Calderon-Zygmund decomposition show that the sets

$$A_k := \{f \ge C^k\} \cap Q_1,$$

satisfy $|A_{k+1}| \le \frac{1}{2}|A_k|$, where C is a large constant depending only on n.

c) (John-Nirenberg lemma) Show that

$$\fint_{Q_1} e^{\frac{c_0}{M}|u - \fint_{Q_1} u|} dx \le C,$$

with c_0, C depending only on n.

d) Show that if $f \in W^{1,n}$ then $f \in BMO$ and $[f]_{BMO} \le C_n \|\nabla f\|_{L^n}$.

9. The H^1 theory

(Sections 8.1-8.4 in Gilbarg-Trudinger, Chapter 6 in Evans)

9.1. **Laplace equation.** There are two methods of proving uniqueness for

$$\triangle u = 0 \quad \text{in } \Omega, \quad u = 0 \quad \text{on } \partial\Omega.$$

Here, for simplicity we are assuming that $u \in C^2(\overline{\Omega})$ and say Ω is a bounded C^1 domain.

Method 1. If u has some positive values then we can touch u by above with a multiple of $R^2 - |x|^2$, say at some $x_0 \in \Omega$. We obtain $D^2u(x_0) < 0$ and contradict the equation. This is a general argument, and it applies for equations

$$tr(A(x)D^2u) = a_{ij}(x)u_{ij} = 0$$

provided that the matrix coefficients $A(x) \geq 0$ is symmetric and $A(x)$ has one positive eigenvalue, but no continuity in x is required. This method leads to the theory of viscosity solutions for second order elliptic equations.

Method 2. We multiply the equation by u and integrate by parts by writing $\triangle u = div(\nabla u)$. We obtain

$$0 = \int_\Omega (\triangle u)u\,dx = -\int_\Omega |\nabla u|^2 dx,$$

hence u is constant in each connected component. This argument applies for equations of the type

$$div(A(x)\nabla u) = div(a_{ij}(x)u_j) = 0,$$

with $a_{ij}(x)\xi_i\xi_j > 0$ for any unit direction ξ, and a_{ij} measurable in x.

Here we used the fact that there is a natural energy $\int |\nabla u|^2 dx$ (called Dirichlet energy) associated to Laplace equation. This method leads to the L^2 theory (or energy estimates) for divergence second order equations.

The Dirichlet problem in H^1

We look for a function $u \in H^1_0(B_1)$ that solves

$$\triangle u = f \quad \text{in } B_1, \tag{9.1}$$

where $f \in L^2(B_1)$ is given. The equation above is understood in the weak sense, that is, by definition (9.1) means

$$\int_{B_1} \nabla u \cdot \nabla \varphi + f\varphi\,dx = 0, \qquad \forall\,\varphi \in C_0^\infty(B_1).$$

By approximation the equality holds for all $\varphi \in H^1_0(B_1)$. So we look for $u \in H^1_0(B_1)$ such that

$$u \cdot \varphi = -\int f\varphi\,dx \quad \forall\,\varphi \in H^1_0(B_1),$$

where $u \cdot \varphi$ represents the inner product in the Hilbert space $H^1_0(B_1)$.

The existence of u follows from Riesz representation theorem since the right hand side is a bounded linear function $L(\varphi)$ in $H_0^1(B_1)$

$$|L(\varphi)| \le \|f\|_{L^2}\|\varphi\|_{L^2} \le \|f\|_{L^2}\|\varphi\|_{H_0^1}.$$

Equivalently, the existence of u follows by minimizing the convex function

$$F(u) := \int_{B_1} \frac{1}{2}|\nabla u|^2 + fu\,dx, \qquad u \in H_0^1(B_1).$$

We pick a sequence u_k so that $F(u_k) \to \inf F$. Notice that u_k is a bounded sequence in H_0^1 since $F(u) \to \infty$ as $\|u\| \to \infty$. We can extract a weakly convergent subsequence, say $u_k \rightharpoonup u$. Then the limit u is a minimizer of F (and a solution to our problem) since by convexity of F we have

$$\liminf F(u_k) \ge F(u).$$

These arguments apply also if the right hand side f is of the form $f + div\, g$ with $g = (g_1, g_2, .., g_n) \in L^2$. We summarize this discussion.

Theorem 9.1 (Solvability of the Dirichlet problem in H^1)**.** *There exists a unique function $u \in H^1(B_1)$ such that*

$$\triangle u = f + \partial_i g_i \quad \text{in } B_1, \qquad u - \varphi \in H_0^1(B_1),$$

where f, $g_i \in L^2$ and $\varphi \in H^1(B_1)$. Moreover,

$$\|u\|_{H^1} \le C_n\ (\|f\|_{L^2} + \sum \|g_i\|_{L^2} + \|\varphi\|_{H^1}).$$

By definition, the equation above means that

$$\int_{B_1} \nabla u \cdot \nabla v + fv - g \cdot \nabla v\,dx = 0, \qquad \forall\, v \in H_0^1(B_1). \tag{9.2}$$

The space of right hand sides $f + div\, g$ with f, $g \in L^2$ is called H^{-1} and it acts (linearly) on H_0^1 functions as

$$< f + div\, g, u >:= \int fu - g \cdot \nabla u\,dx.$$

Thus

$$H^{-1} = (H_0^1)^*, \qquad \|f + div\, g\|_{H^{-1}} \le \|f\|_{L^2} + \|g\|_{L^2}.$$

Fredholm alternative. The theorem above shows that $\triangle : H_0^1(B_1) \to H^{-1}(B_1)$ is an isomorphism, hence $\triangle^{-1} : H^{-1} \to H_0^1$ is bounded. Since H_0^1 is compactly embedded in $L^2 \subset H^{-1}$, we see that $\triangle^{-1}$ is a compact operator when viewed from $H^{-1} \to H^{-1}$ or $L^2 \to L^2$. In particular the equation

$$\triangle u + \lambda u = f \in L^2, \qquad u \in H_0^1,$$

can be written as

$$(I + \lambda T)u = Tf, \quad T : L^2 \to L^2 \quad \text{compact, selfadoint}$$

and this is solvable if and only if

$$Tf \in range(I + \lambda T) = ker(I + \lambda T)^{\perp}$$

which means

$$\int fv\,dx = 0 \quad \text{for all} \quad v \in V_\lambda := \{v \in H_0^1 | \quad \triangle v + \lambda v = 0\}.$$

The eigenspace V_λ is finite dimensional and is nontrivial only for s sequence of $\lambda \to \infty$.

Caccioppoli inequality. *If*

$$\triangle u = f + div\, g \quad in\ B_1,$$

then

$$\|\nabla u\|_{L^2(B_{1/2})} \le C_n \left(\|u\|_{L^2(B_1)} + \|f\|_{L^2(B_1)} + \|g\|_{L^2(B_1)} \right).$$

We take $v = \eta^2 u$ as test function in (9.2), with $\eta \in C_0^\infty(B_1)$ and use

$$\eta \nabla u = \nabla(\eta u) - u \nabla \eta, \qquad ab \le \delta a^2 + \delta^{-1} b^2,$$

and obtain

$$\int_{B_1} |\nabla(\eta u)|^2 \le C \int |\nabla \eta|^2 u^2 + \eta^2 u^2 + f^2 + |g|^2\, dx.$$

This gives the desired inequality by choosing $\eta = 1$ in $B_{1/2}$.

We also mention the boundary version when u vanishes (in the trace sense) on the boundary :

If

$$\triangle u = f + div\, g \quad in\ B_1^+, \qquad u = 0 \quad on\ \{x_n = 0\},$$

then

$$\|\nabla u\|_{L^2(B_{1/2}^+)} \le C_n \left(\|u\|_{L^2(B_1^+)} + \|f\|_{L^2(B_1^+)} + \|g\|_{L^2(B_1^+)} \right).$$

Higher regularity. *Assume*

$$\triangle u = f \quad in\ B_1.$$

If $f \in L^2$ then $u \in H^2_{loc}(B_1)$ and

$$\|u\|_{H^2(B_{1/2})} \le C \left(\|u\|_{L^2(B_1)} + \|f\|_{L^2(B_1)} \right).$$

If $f \in H^k$ then $u \in H^{k+2}_{loc}(B_1)$ and

$$\|u\|_{H^{k+2}(B_{1/2})} \le C \left(\|u\|_{L^2(B_1)} + \|f\|_{H^k(B_1)} \right).$$

Indeed, by differentiating in the e_i direction we obtain

$$\triangle u_i = \partial_i f \in H^{-1},$$

and by Caccioppoli inequality we can bound $\|u_i\|_{H^1_{loc}}$ in terms of $\|u\|_{H^1}$ and $\|f\|_{L^2}$. This implies $u \in H^2_{loc}$. This argument is formal since we cannot write the equation above for u_i, we do not know $u_i \in H^1$. However we can use the usual trick of working with discrete differences instead and obtain the desired estimates in the limit.

The higher estimates follow by induction on k.

These estimates show that weak solutions to $\triangle u = 0$ satisfy $u \in H^k_{loc}$ for all $k \ge 0$, therefore they are classical solutions $u \in C^\infty_{loc}$ by Sobolev embedding.

A boundary version of the result above is stated in Problem 1 below.

9.2. **General divergence equations.** The results above for $\triangle$ can be extended to general divergence operators involving lower order terms

$$Lu := \partial_i \left(a^{ij}(x)\, u_j + b^i(x)\, u\right) + c^i(x)\, u_i + d(x)\, u,$$

with measurable coefficients satisfying

$$\Lambda|\xi^2| \ge a^{ij}(x)\, \xi_i\xi_j \ge \lambda|\xi|^2, \quad \lambda > 0,$$

and b^i, c^i, d bounded (or in appropriate L^p spaces). By definition, u solves

$$Lu = f + div\, g \quad \text{in } \Omega,$$

means that

$$\int a^{ij}\, u_j\varphi_i + b^i\, u\, \varphi_i - c^i u_i\varphi - d\, u\, \varphi + f\varphi - g^i\varphi_i \; dx = 0, \qquad \forall \varphi \in C_0^\infty(\Omega).$$

When $u \in H^1(\Omega)$ then the equality above holds for all $\varphi \in H_0^1(\Omega)$.

For simplicity we discuss the case when

$$Lu := \partial_i(a^{ij}(x)u_j), \tag{9.3}$$

although the same analysis carries to the general case (see Problem 2). We only remark that the maximum principle, existence and uniqueness hold when we have lower order terms provided that the usual sign condition holds in the weak sense for the coefficient corresponding to the zero order term

$$div\, b + d \le 0.$$

Maximum principle. *Assume that $Lu \ge 0$, with L as in (9.3) and $u \in H^1(\Omega)$. If $u \le 0$ on $\partial\Omega$ (in the sense that $u^+ \in H_0^1(\Omega)$) then $u \le 0$ in Ω.*

Indeed, we take as test function $\varphi = u^+ \in H_0^1(\Omega)$ and obtain

$$0 \ge \int_\Omega a^{ij}\, u_j\, (u^+)_i\, dx = \int_\Omega a^{ij}\, (u^+)_j\, (u^+)_i\, dx \ge \lambda \int |\nabla u^+|^2\, dx,$$

which implies $u^+ = 0$.

Dirichlet problem. *There exists a unique function $u \in H^1(\Omega)$ such that*

$$Lu = f + \partial_i g_i \quad \text{in } \Omega, \qquad u - \varphi \in H_0^1(\Omega),$$

where f, $g_i \in L^2$ and $\varphi \in H^1(\Omega)$. Moreover,

$$\|u\|_{H^1} \le C\, (\|f\|_{L^2} + \sum \|g_i\|_{L^2} + \|\varphi\|_{H^1}).$$

with C a constant depending on λ, Λ, n.

If $A := (a^{ij})$ is symmetric then we can obtain the solution by minimizing the energy

$$E(u) = \int_\Omega \frac{1}{2}\, \nabla u\; A\, (\nabla u)^T + fu - g \cdot \nabla u\, dx, \qquad u - \varphi \in H_0^1(\Omega).$$

If A is not symmetric then we need to use the Lax-Milgram Theorem which is an extension of the Riesz representation theorem to the setting of non-symmetric bilinear forms:

Let $B(u, v)$ be a bilinear form on a Hilbert space X, that satisfies

$$|B(u, v)| \le C|u|\, |v|, \qquad B(u, u) \ge c|u|^2,$$

for some positive constants c, C. Then, given a bounded linear function L on X we can find a unique u such that $B(u,v) = L(v)$ for all $v \in X$.

For a proof of the Lax-Milgram Theorem see Problem 2 below.

In our case, we reduce to the case $\varphi = 0$, and then we apply the theorem above with $X = H_0^1(\Omega)$,

$$B(u,v) := \int_\Omega \nabla v \, A \, (\nabla u)^T \, dx, \qquad L(v) = \int_\Omega g \cdot \nabla v - f v \, dx.$$

Fredholm alternative. As above $L^{-1} : L^2 \to L^2$ is compact, and the equation

$$Lu + \lambda u = f \in L^2,$$

is solvable in H^1 if and only if

$$\int f v \, dx = 0 \quad \text{for all} \quad v \in V_\lambda := \{v \in H_0^1 | \quad L^* v + \lambda v = 0\},$$

where $L^* v := \partial_i (a^{ji} \partial_j v)$ is the adjoint of L (that is A is replaced by A^T).

Caccioppoli inequality. The statement remains the same. *If*

$$Lu = f + div \, g \quad in \ B_1,$$

then

$$\|\nabla u\|_{L^2(B_{1/2})} \leq C \left(\|u\|_{L^2(B_1)} + \|f\|_{L^2(B_1)} + \|g\|_{L^2(B_1)} \right).$$

Higher regularity. Up to know we did not require any continuity on the coefficients $a^{ij}(x)$. However for higher regularity we need to assume they are Lipschitz.

Suppose

$$Lu = f \quad in \ B_1,$$

and a^{ij} are Lipschitz continuous. If $f \in L^2$ then $u \in H^2_{loc}(B_1)$ and

$$\|u\|_{H^2(B_{1/2})} \leq C \left(\|u\|_{L^2(B_1)} + \|f\|_{L^2(B_1)} \right).$$

Moreover, if $a^{ij} \in C^{k,1}$ and $f \in H^k$ then $u \in H^{k+2}_{loc}(B_1)$ and

$$\|u\|_{H^{k+2}(B_{1/2})} \leq C \left(\|u\|_{L^2(B_1)} + \|f\|_{H^k(B_1)} \right).$$

Indeed, by differentiating formally in the lth direction we find that u_l solves

$$L \, u_l = \partial_l f - \partial_i (a_l^{ij} u_j),$$

and Caccioppoli inequality implies the desired estimates.

Similarly, the higher derivatives estimates hold up to the boundary provided that $\partial\Omega \in C^{1,1}$, or $\partial\Omega \in C^{k+1,1}$ for $k \geq 1$.

9.3. **Change of variables.** Divergence equations preserve naturally when changing variables. Assume that $u(x)$ solves

$$Lu = 0 \quad \Longleftrightarrow \quad \int \nabla v \, A(x)(\nabla u)^T \, dx = 0 \quad \text{in } \Omega_x.$$

Let

$$\Phi : \Omega_x \to \Omega_y, \qquad y = \Phi(x),$$

be a bi-Lipschitz map between the domains Ω_x and Ω_y. We write for simplicity $D_x y$ instead of $D\Phi(x)$ and $D_y x$ for $\Psi^{-1}(y)$ for the change of variables matrices between the x and y variables. We write the equation for u when viewed as a function of y. We use that

$$\nabla_x u = \nabla_y u \; D_x Y$$

and then

$$\begin{aligned} 0 &= \int \nabla_x v \, A(x) \, (\nabla_x u)^T \, dx \\ &= \int (D_x y \, \nabla_y v) \, A(x) \, (\nabla_y u \, D_x y)^T \, |D_y x| dy \\ &= \int \nabla_y v \, \overline{A}(y) \, (\nabla_y u)^T \, dy, \qquad \overline{A}(y) := D_x y \, A(x) \, (D_x y)^T \; |D_y x|. \end{aligned}$$

Thus in the y variable u solves

$$\bar{L}u = 0 \quad \text{in } \Omega_y, \quad \text{with} \quad \overline{L}u = div(\bar{A}(y) \, \nabla_y u),$$

with $\overline{A}$ as above. Notice that $\overline{L}$ is uniformly elliptic as well.

In conclusion any bi-Lipschitz deformation of a solution to a divergence equation (for example a harmonic function) solves a similar divergence equation.

9.4. **Schauder estimates for divergence equations.** *Assume that*

$$Lu = f + div \; g \qquad in \; B_1,$$

with $a^{ij} \in C^\alpha$, $g \in C^\alpha$, *for some* $\alpha \in (0,1)$ *and* $f \in L^p$ *with* $\alpha \le 1 - \frac{n}{p}$. *Then* $u \in C^{1,\alpha}(B_1)$ *and*

$$\|u\|_{C^{1,\alpha}(B_{1/2})} \le C \left(\|u\|_{L^2} + \|f\|_{L^p} + \|g\|_{C^\alpha} \right).$$

Similar estimates hold up to the boundary if the boundary data and the domain are of class $C^{1,\alpha}$.

The idea is use Campanoto approach. We only sketch the main ideas.

Assume that $a^{ij}(0) = \delta_i^j$, $g(0) = 0$, $\|u\|_{L^2}$, $\|f\|_{L^p}$, $\|g\|_{C^\alpha}$ are sufficiently small and then show by induction that for a sequence of radii $r = \rho^k$ we have bounded linear functions l_r such that

$$⨍_{B_r} (u - l_r)^2 dx \le r^{2+2\alpha}.$$

Notice that the rescaled function $\tilde{u} = r^{-(1+\alpha)}(u - l_r)(rx)$ satisfies $\|u\|_{L^2(B_1)} \le C$, and satisfies the equation

$$\tilde{L}\tilde{u} := \partial_i(\tilde{a}^{ij}(x)\tilde{u}_j) = \tilde{f} + div(\tilde{g}),$$

with

$$\tilde{a}^{ij}(x) = a^{ij}(rx), \quad \tilde{f}(x) = r^{1-\alpha} f(rx), \qquad \tilde{g}(x) = r^{-\alpha}[g - (a^{ij} - \delta_j^i)\partial_j l_r](rx).$$

Notice that $\|\tilde f\|_{L^p}$ and $\|\tilde g\|_{L^\infty}$ are small, and the Caccioppoli inequality implies $\|\tilde u\|_{H^1(B_{1/2})}$ is bounded. Now we compare $\tilde u$ with the harmonic function w with the same boundary data on $\partial B_{1/2}$ and obtain that $\tilde u - w \in H^1_0(B_{1/2})$ solves

$$\tilde L(u-w) = \tilde f + div\,\bar g, \qquad \bar g := \tilde g - (\tilde A - I)\nabla w.$$

Since $\tilde f$ and $\bar g$ have small L^2 norms we find that $\|\tilde u - w\|_{H^1_0}$ is sufficiently small which by the C^2 estimates for w at the origin implies

$$\fint_{B_\rho} (\tilde u - l_\rho)^2 dx \le \rho^{2+2\alpha}.$$

9.5. **Problems.**

1. Assume $u \in H^1_{loc}(B_1^+)$ vanishes in the trace sense on $\{x_n = 0\}$,

$$\triangle u = f \quad \text{in } B_1^+, \qquad\qquad u = 0 \quad \text{on } \{x_n = 0\}.$$

If $f \in L^2$ then $u \in H^2_{loc}(B_1^+)$ and

$$\|u\|_{H^2(B^+_{1/2})} \le C\left(\|u\|_{L^2(B_1^+)} + \|f\|_{L^2(B_1^+)}\right).$$

If $f \in H^k$ then $u \in H^{k+2}_{loc}(B_1)$ and

$$\|u\|_{H^{k+2}(B^+_{1/2})} \le C\left(\|u\|_{L^2(B_1^+)} + \|f\|_{H^k(B_1^+)}\right).$$

2. a) Assume that X is a Hilbert space and $T : X \to X$ (not necessarily linear) satisfies

$$(T(x) - T(y), x - y) \ge c|x-y|^2, \quad |T(x) - T(y)| \le C|x-y|.$$

Prove that T is 1-1 by showing that $I - \delta T$ is a contraction if δ is sufficiently small.

b) Deduce the Lax-Milgram Theorem.

3. a) Assume that

$$Lu := \partial_i\left(a^{ij}(x)\,u_j + b^i(x)\,u\right) + c^i(x)\,u_i + d(x)\,u = f + \partial_i g^i.$$

Show that $\tilde u(x) := u(rx)$ with r small solves a similar equation

$$\tilde L \tilde u = \tilde f + \partial_i \tilde g^i,$$

and identify the correct L^p spaces for each of the coefficients b, c, d, f, g so that their corresponding norms improved in the rescaled equation.

b) Show that Caccioppoli inequality holds in this setting.

c) Show that the maximum principle holds if $divb + d \le 0$ in the weak sense.
(Hint: use as test function $(u-m)^+$ with m such that its gradient is nonzero on a set of small positive measure)

d) Check that the Fredholm alternative holds for L as in c).

4. (*Serrin's example*)
Verify that $u(x) = x_1|x|^{-1-\varepsilon}$ is a weak solution (in the sense of distributions) to a uniformlly elliptic equation in 2D

$$div(A\,\nabla u) = 0$$

for some A of the form

$$A(x) := \delta_i^j + a\frac{x_i x_j}{|x|^2},$$

with $a \geq 0$, depending on ε. Deduce that Caccioppoli inequality does not hold without the assumption that $u \in H^1(B_1)$.

5. *Eigenvalues of the Laplace operator.* (see Evans Chapter 6)
Let Ω be a bounded C^2 domain let

$$\mathcal{A} := \{u \in H_0^1(\Omega)| \quad \|u\|_{L^2} = 1\}.$$

a) Define

$$\lambda_1 := \inf_{u\in\mathcal{A}} \int_\Omega |\nabla u|^2 dx.$$

Show that the infimum is achieved by a function w_1 which solves

$$\triangle w_1 = -\lambda_1 w_1.$$

b) Show that w_1 is smooth in Ω, and if $(w_1)^+$ is not identically 0 then a multiple of $(w_1)^+$ achives the infimum λ_1 as well. Conclude that w_1 must be positive and unique (up to a sign change), thus λ_1 is a simple eigenvalue.

c) Define inductively eigenfunctions $w_k \subset H_0^1$ by selecting a minimizer of the problem

$$\inf_{u\in\mathcal{A}\cap w_1^\perp\cap\ldots\cap w_{k-1}^\perp} \int_\Omega |\nabla u|^2 dx =: \lambda_k.$$

where $w_i^\perp$ denotes the subspace of functions orthogonal to w_i with the L^2 inner product.
Show that $\lambda_k \to \infty$, and that $\{w_k\}$ form an orthonormal basis for both $H_0^1(\Omega)$ and $L^2(\Omega)$.

6. a) Let λ_k be the eigenvalues of the Laplace operator in $H_0^1(\Omega)$ as above. Show that

$$\lambda_k = \min_{\Sigma_k} \max_{u\in\Sigma_k\setminus\{0\}} \frac{\int|\nabla u|^2}{\int u^2},$$

where Σ_k represents a k-dimensional vector subspace of $H_0^1(\Omega)$.

b) For a domain Ω and $t \geq 0$ we denote by

$$\lambda_\Omega(t) = \{k| \quad \lambda_k \leq t, \quad \lambda_{k+1} > t\},$$

the number of eigenvalues less than t. Then

$$\lambda_\Omega(t) = \max\left\{k|\quad \exists\,\Sigma_k \subset H^1_0(\Omega) \quad \text{such that} \quad \int |\nabla u|^2 \le t\int u^2, \quad \forall u \in \Sigma_k\right\},$$

Deduce that $\lambda_\Omega(t) \le \lambda_U(t)$ if $\Omega \subset U$.

d) Let μ_k denote the eigenvalues of Laplace operator in $H^1(\Omega)$ (the Neumann eigenvalues). Deduce parts a), b) in this setting and denote by $\mu_\Omega(t)$ the corresponding counting function of eigenvalues. Assume that Ω is divided by a Lipschitz surface into two disjoint domains Ω_1 and Ω_2. Show that

$$\lambda_{\Omega_1}(t) + \lambda_{\Omega_2}(t) \le \lambda_\Omega(t) \le \mu_\Omega(t) \le \mu_{\Omega_1}(t) + \mu_{\Omega_2}(t).$$

e) Use that for a cube Q, as $t \to \infty$, $t^{-n/2}\lambda_Q(t)$ and $t^{-n/2}\mu_Q(t)$ converge to $c_n|Q|$ for some explicit constant c_n. Deduce from part d) the asymptotic Weyl's law

$$\lim_{t\to\infty} \frac{\lambda_\Omega(t)}{t^{\frac{n}{2}}} = c_n|\Omega|.$$

7. Let Ω be a bounded C^2 domain and assume that $u \in C^2(\Omega)\cap C^0(\overline{\Omega})$ satisfies $u > 0$ in Ω and solves

$$\triangle u = -f(x)u \quad \text{in } \Omega, \quad u = 0 \quad \text{on } \partial\Omega.$$

Show that any solution $v \in C^2(\Omega)\cap C^0(\overline{\Omega})$ to $\triangle v = -g(x)v$ with $g \ge f$ must either change sign or coincide with a multiple of u. On the other hand if $g \le f$ and $g \ne f$ we can find a strictly positive solution $v > 0$ in $\overline{\Omega}$.

10. De Giorgi - Nash - Moser theorem

(Sections 8.5 - 8.8 in [GT], Chapter 4 in Han- Lin [HL])

The results of this section apply to general uniformly elliptic operators

$$Lu = \partial_i(a^{ij}u_j) + b^i u_i + cu,$$

with

$$\Lambda|\xi^2| \geq a^{ij}\xi_i\xi_j \geq \lambda|\xi|^2, \qquad b \in L^q, \quad c \in L^{q/2} \quad \text{for some} \quad q > n,$$

and weak solutions (or subsolutions/supersolutions) $u \in H^1(B_1)$ to

$$Lu = f + div\, g \qquad \text{in } B_1, \qquad f \in L^{q/2}, \quad g \in L^q.$$

To focus on the main ideas, we assume

$$b,\ c,\ f,\ g \text{ are identically } 0.$$

In the general case, the constants below will depend also on the norms of b and c, and on the right hand sides we need to add the norms of f and g next to the norms of u.

Theorem 10.1. *Assume that*

$$Lu \geq 0 \quad \text{in } B_1.$$

Then

$$u \leq C\, \|u^+\|_{L^2(B_1)} \quad \text{in } B_{1/2},$$

with C depending on n, λ, Λ.

Recall that $u^+ = \max\{u, 0\}$.

Proof. After multiplication by a constant we may assume $\|u\|_{L^2}$ is sufficiently small, and we need to show that $u \leq 1$ in $B_{1/2}$.

We use as test function $v = \eta^2 u^+$ and Caccioppoli inequality gives

$$\int |\nabla(\eta u^+)|^2 dx \leq C \int |\nabla \eta|^2 (u^+)^2 dx.$$

By Sobolev embedding and Holder inequality

$$\int |\nabla(\eta u^+)|^2 dx \geq \|\eta u^+\|^2_{L^{2^*}} \geq \|\eta u^+\|^2_{L^2} |\{\eta u^+ > 0\}|^{-2/n}.$$

In conclusion

$$\int (\eta u^+)|^2 dx \leq C|\{\eta u^+ > 0\}|^{\frac{2}{n}} \int |\nabla \eta|^2 (u^+)^2 dx,$$

and this inequality holds if we replace u^+ by $(u-t)^+$.

Denote by

$$a_k := \int_{A_k} (u - t_k)^2 dx,$$

with

$$t_k := 1 - 2^{-k}, \qquad A_k := |\{u > t_k\} \cap B_{2^{-1}+2^{-k}}|\,,$$

and it suffices to show that $a_k \to 0$ as $k \to \infty$.

In the the inequality above for $(u - t_k)^+$, we choose η_k to be supported in $B_{2^{-1}+2^{-k}}$ and be 1 in $B_{2^{-1}+2^{-(k+1)}}$, and obtain

$$a_{k+1} \leq C2^{4k}\, a_{k-1}^{\frac{2}{n}}\, a_k \leq C2^{4k}\, a_{k-1}^{1+\frac{2}{n}},$$

where we used that

$$|A_k| \leq 2^{2k} a_{k-1}, \qquad |\nabla \eta_k| \leq C2^k.$$

Now it follows easily that $a_k \to 0$ if

$$a_{k+1} \leq C2^{4k} a_{k-1},$$

and a_0 is sufficiently small. □

Theorem 10.2 (Weak Harnack inequality). *Assume that*

$$Lu \geq 0 \quad in\ B_1.$$

If

$$|\{u \leq 0\} \cap B_1| \geq \delta |B_1|,$$

for some $\delta > 0$ then

$$u^+ \leq (1 - c(\delta))\|u^+\|_{L^\infty(B_1)} \quad in\ B_{1/2},$$

with $c(\delta) > 0$ small, depending on n, λ, Λ and δ.

Proof. If $\{u \leq 0\}$ almost covers B_1 in measure, then $\|u^+\|_{L^2}$ is small and the result follows since $u \leq 1/2$ in $B_{1/2}$ by Theorem 10.1.

It suffices to show that $\{u > t_k\}$, $t_k = 1 - 2^{-k}$, has small measure in B_1 for some large k, and we apply the reason above for

$$v := \frac{(u - t_k)^+}{1 - t_k}.$$

For this we show that the set $\{t_k < u < t_{k+1}\} = \{0 < v < 1/2\}$ cannot have very small measure and $\{v \leq 0\}$ and $\{v \geq 1/2\}$ have measure bounded by below. Otherwise we contradict Caccioppoli inequality for v,

$$\int_{B_{1-\sigma}} |\nabla v|^2 dx \leq C(\sigma) \int v^2 \leq C(\sigma).$$

since the left hand side becomes very large if $\{0 < v < 1/2\}$ has small measure. This follows by Holder's inequality and the Poincare inequality which implies

$$\int_{B_{1-\sigma} \cap \{0 < v < 1/2\}} |\nabla v| \ dx \geq c.$$

□

Remark 10.3. We remark that the hypothesis that $\{u \leq 0\}$ has at least δ measure in B_1 can be replaced to $u \leq 0$ on $\partial B_1 \cap B_\delta(x_0)$ for some $x_0 \in \partial B_1$ (in the sense of traces). Indeed, we work with u^+ which can be extended to 0 in $B_\delta(x_0)$ outside B_1. Then u^+ satisfies the hypotheses of the Theorem in the extended domain.

As a corollary of Theorems 10.1 and 10.2 we obtain the Holder continuity of solutions to $Lu = 0$.

Theorem 10.4 (Holder continuity)**.** *Assume that $u \in H^1(B_1)$ solves*

$$Lu = 0 \quad in\ B_1.$$

Then $u \in C^\alpha(B_1)$, and

$$\|u\|_{C^\alpha(B_{1/2})} \le C\|u\|_{L^2},$$

where α, C are constants depending on n, λ, Λ.

Proof. By Theorem 10.1 we know that u is bounded in $B_{3/4}$. To prove the Holder continuity it suffices to show that

$$osc\ _{B_{r/2}} u \le (1-c)\ osc_{B_r} u.$$

This statement is invariant under rescalings so we may assume that $r = 1$, and $-1 \le u \le 1$ in B_1. We want to show that when we restrict to $B_{1/2}$ we can improve one of the two bounds of u. This follows from Theorem 10.2 applied either to u^+ or u^- depending whether or not $\{u \le 0\}$ covers or not more than half of B_1.

□

From Remark 10.3 it follows that if $u = \varphi$ on ∂B_1 and $\varphi \in C^\alpha$ then we obtain the global version of Theorem 10.4, i.e. $u \in C^\alpha(\overline{B}_1)$.

Next we show that weak Harnack inequality implies the following Harnack inequality for solutions.

Theorem 10.5 (Harnack inequality)**.** *Assume that*

$$Lu = 0 \quad and \quad u \ge 0 \quad in\ B_1.$$

Then

$$u \le C\ u(0) \quad in\ B_{1/2},$$

with C depending on λ, Λ, n.

We remark that if we solve $Lu = f + div\ g$ then $u(0)$ in the theorem above needs to be replaced by $u(0) + \|f\|_{L^{q/2}} + \|g\|_{L^q}$.

Next we prove some equivalent versions of Weak Harnack inequality. The starting point is that we assume Theorem 10.2 holds. Then if u is a nonnegative supersolution with $u(0) = 1$, then $(C-u)^+$ which is a subsolution and obtain

Weak Harnack inequality (supersolution version) Assume that

$$Lu \le 0 \quad \text{and} \quad u \ge 0 \quad \text{in } B_1,$$

Then

$$|\{u > C(\delta)\ u(0)\} \cap B_1| \le \delta |B_1|,$$

if $C(\delta)$ is large, depending on n, λ, Λ, δ.

This statement (for some fixed δ_0 small) can be iterated by Calderon-Zygmund decomposition and obtain a decay of the distribution function

$$a(t) := |\{u > t\} \cap B_{1/2}|.$$

Precisely, we want to show that if C is chosen large, then

$$a(C\,t) \le \frac{1}{2}\ a(t). \tag{10.1}$$

After diving by t we may assume $t = 1$. For simplicity we switch to cubes instead of balls and we assume that $u \geq 0$, $Lu \leq 0$ in Q_2 the cube of size 2. We decompose the set

$$A := \{u > C\} \cap Q_1$$

into disjoint dyadic cubes Q_i such that the density of A into each Q_i is below $1/2$ but it is above $1/2$ for one of the next dyadic subcubes Q_i' of Q_i,

$$|Q_i \cap A| \leq \frac{1}{2}|Q_i|, \qquad |Q_i' \cap A| \geq \frac{1}{2}|Q_i'|.$$

Then the weak Harnack inequality for supersolutions implies that $Q_i \subset \{u > 1\}$, (by choosing δ sufficiently small) which gives the desired claim (10.1) after summing over all cubes Q_i.

In conclusion we obtain that if $u(0) = 1$ then $a(t) \leq Ct^{-M}$ for all $t \geq 1$, with M large which implies

Weak Harnack inequality (supersolution version 2) Assume that

$$Lu \leq 0 \quad \text{and} \quad u \geq 0 \quad \text{in } B_1, \text{ and} \quad u(0) = 1.$$

Then

$$\int_{B_{1/2}} u^p \, dx \leq C, \tag{10.2}$$

if C is large, and $p > 0$ small depending on n, λ, Λ.

Remarks:

1) It turns out that the range of p in this statement can be improved for all $p < \frac{n}{n-2}$ which is optimal even for Laplace equation, see Problem 2.

2) This statement gives a more precise dependence of the constants $c(\delta) \sim \delta^M$ and $C(\delta) \sim \delta^{-M}$ as $\delta \to 0$, in the statements of weak Harnack inequalities above.

Finally we give a similar integral version of the weak Harnack inequality for subsolutions which improves slightly the statement of Theorem 10.1.

Weak Harnack inequality (subsolution version 2) Assume that

$$Lu \geq 0 \quad \text{in } B_1.$$

Given any $p > 0$, there exists $C(p)$ depending on p, n, λ, Λ such that

$$u(0) \leq C \left(\int_{B_1} (u^+)^p \, dx \right)^{\frac{1}{p}}. \tag{10.3}$$

Proof. Suppose $u \geq 0$ and $\|u\|_{L^p} = 1$, and assume by contradiction that $u(0) \gg 1$. We construct inductively a sequence of points y_k, $y_0 = 0$, such that

$$u(y_{k+1}) \geq (1 + c_0)u(y_k), \qquad |y_{k+1} - y_k| \leq C \; u(y_k)^{-\sigma},$$

with C, c_0, σ universal constants. Then y_k is convergent and we contradict that u must be bounded in $B_{1/2}$, see Theorem 10.1.

Let r be the largest radius so that

$$u \leq (1 + c_0)u(y_k) \quad \text{ in } B_r(y_k).$$

The weak Harnack inequality for supersolutions applied to

$$v = (1 + c_0)u(y_k) - u$$

shows that $u \geq \frac{1}{2}u(u_k)$ in a fixed proportion of $B_r(y_k)$. Thus

$$1 \geq \int_{B_r(y_k)} u^p\, dx \geq cr^n\, u^p(y_k),$$

which gives

$$r \leq Cu(y_k)^{-\sigma},$$

and our claim is proved.

□

Now Harnack inequality Theorem 10.5 follows easily from the integral versions of the Weak Harnack inequality for subsolutions and supersolutions.

10.1. **Problems.**

1. Prove Theorem 10.1 by taking as test functions

$$v = \eta_k\ (u^+)^{2\beta_k - 1},$$

with $\beta_0 = 1$, and $\beta_{k+1} = \beta_k 2^*/2$, and obtain an inequality

$$a_{k+1} \leq C(k)\, a_k, \qquad a_k := \|u^+\|_{L^{2\beta_k}(B_{2^{-1}+2^{-k}})},$$

and conclude that a_k remains bounded as $k \to \infty$.

2. Assume that

$$Lu \leq 0 \quad \text{and} \quad u > 0 \quad \text{in } B_1.$$

Use as test functions the functions v as above with $\beta \in (0, 1/2)$ and conclude that the weak Harnack inequality (10.2) holds for $p < 2^*/2$. Show that this is optimal by considering fundamenetal solutions of the Laplace equation.

What happens when β tends to 0 ?

3. Show that if

$$Lu = \partial_i(a^{ij}u_j) = 0 \quad \text{in } \Omega,$$

then $L(\phi(u)) \geq 0$ for any $\phi : \mathbb{R} \to \mathbb{R}$ convex.

4.(Boundary Harnack inequality) Let

$$Lu := \partial_i(a_{ij}\partial_j u),$$

be a uniformlly elliptic operator in B_1^+. Assume that

$$u \geq 0 \text{ and } Lu = 0 \text{ in } B_1^+,$$

and

$$u = 0 \text{ on } \{x_n = 0\},\ u(1/2e_n) = 1.$$

Using Harnack inequality show that

$$\int_{B_{1/2}^+} u^\delta \le C,$$

and deduce from the Weak Harnack inequality that $u \le C$ in $B_{1/2}^+$.

5. (The Green function) Let w_ρ be the solution to

$$Lw_\rho = -\frac{1}{|B_\rho(y)|}\chi_{B_\rho(y)}, \qquad w_\rho = 0 \quad \text{on } \partial B_1,$$

for some $y \in B_{1/2}$ and ρ small.

a) By using Harnack inequality show that the values of w_ρ are comparable on the diadic regions $D_r := B_{2r}(y) \setminus B_{r/2}(y)$ with $r \ge \rho$ around y:

$$\max_{D_r} w_\rho / \min_{D_r} w_\rho \le C(n, \lambda, \Lambda).$$

b) Prove that if $0 \le s < t \le \min_{B_\rho} w_\rho$ then

$$\int_{s<w_\rho<t} a_{ij}\, \partial_j w_\rho\, \partial_i w_\rho dx = t - s.$$

c) Let $\Omega \subset\ B_1 \setminus B_\rho$ and u denote the minimizer of the Dirichlet energy

$$\int_{B_1} |\nabla u|^2 dx,$$

among all functions u which equal to w_ρ outside Ω. Then

$$c\int_\Omega |\nabla u|^2 dx \le \int_\Omega |\nabla w_\rho|^2 dx \le C\int_\Omega |\nabla u|^2 dx,$$

for some universal constants c, C.

d) Deduce from a), b), c) that $w_\rho \sim t^{2-n}$ on $\partial B_t(y)$.

e) Let $\rho \to 0$ and show that $w_\rho \to G(x,y)$ in $W^{1,p}(B_1)$ for any $p < n/(n-1)$, where $G(x,y)$ (the Green function) satisfies:

$$L\, G(x,y) = -\delta_y.$$

11. The Dirichlet problem for the minimal surface equation

(Chapters 13, 14, 15 in Gilbarg-Trudinger [GT])

In this section we discuss existence of classical solutions $u \in C^2(\overline{B}_1)$ to the Dirichlet problem

$$\begin{cases} div(\nabla F(\nabla u)) = 0 & \text{in } B_1, \\ u = \varphi & \text{on } \partial B_1, \end{cases}$$

where $F \in C^{2,\alpha}(\mathbb{R}^n)$, $D^2 F > 0$, and $\varphi \in C^{2,\alpha}$.

The classical example to have in mind is the minimal surface equation

$$F(p) = (1 + |p|^2)^{\frac{1}{2}}.$$

The existence follows from the following apriori estimate and the method of continuity.

Theorem 11.1. *Assume that $u \in C^{2,\alpha}(\overline{B}_1)$ solves the Dirichlet problem above. Then*

$$\|u\|_{C^{2,\alpha}(\overline{B}_1)} \le C(n, \alpha, F, \|\varphi\|_{C^{2,\alpha}}).$$

Proof. The solution u solves the equation

$$div(\nabla F(\nabla u)) = F_{ij}(\nabla u)\, u_{ij} = 0,$$

and by the Schauder estimates it suffices to prove

$$\|u\|_{C^{1,\alpha}(\overline{B}_1)} \le C(F, \|\varphi\|_{C^2}),$$

for some small $\alpha > 0$ depending on the data. Then the coefficients $F_{ij}(\nabla u) \in C^\alpha$, and by higher order Schauder estimates, the $C^{1,\alpha}$ estimate can be iterated to a $C^{k,\alpha}$ estimate provided that $F \in C^{k,\alpha}$.

Step 1: L^∞ bound for u

$$\min \varphi \le u \le \max \varphi.$$

Indeed, by the strong maximum principle, u cannot have an interior maximum or minimum unless it is constant.

Step 2: L^∞ bound for ∇u on ∂B_1

$$|\nabla u| \le C\|\varphi\|_{C^2} \quad \text{on } \partial B_1.$$

Since $\varphi \in C^2$, at each point $x_0 \in \partial B_1$ we can find two linear functions $l^+_{x_0}$ and $l^-_{x_0}$ which trap the boundary data such that

$$l^-_{x_0} \le \varphi \le l^+_{x_0} \quad \text{on } \partial B_1, \qquad l^-_{x_0} = l^+_{x_0} \quad \text{at } x_0.$$

Since $u - l^\pm_{x_0}$ solve an equation like u but with F translated, by Step 1 we conclude that

$$l^-_{x_0} \le u \le l^+_{x_0} \quad \text{in } B_1,$$

which gives the conclusion.

Step 3: L^∞ bound for ∇u in $\overline{B}_1$.

The derivative u_i solves the divergence equation

$$div(D^2 F(\nabla u) \nabla u_i) = 0 \quad \text{in } B_1, \tag{11.1}$$

and by the maximum principle in divergence form its maximum is on the boundary.

Step 4: Interior $C^{1,\alpha}$ estimates.

Since $|\nabla u| \leq C\|\varphi\|_{C^2}$, we find that

$$\Lambda I \geq D^2F(\nabla u) \geq \lambda I$$

for some $\lambda > 0$, and the equation (11.1) is uniformly elliptic.

The De Giorgi-Nash-Moser Theorem gives

$$\|u_i\|_{C^\alpha(B_{7/8})} \leq C\|u_i\|_{L^\infty}.$$

Step 5: C^α estimates near the boundary for the tangential derivatives

We flatten the boundary by a C^2 deformation (say measure preserving), and then the function u solves in B_1^+ an equation of the type

$$div\left((D(x))^T\nabla F(D(x)\nabla u)\right) = 0 \quad \text{in } B_1^+, \qquad u = \tilde{\varphi} \quad \text{on } \{x_n = 0\}, \tag{11.2}$$

With $D(x) \in C^1$ a matrix with bounded inverse.

After differentiating in the x_i direction, $i < n$, we see that u_i solves an equation of the type

$$div(A(x)\nabla u_i) = div\ g \quad \text{in } B_1^+, \qquad u_i = \tilde{\varphi}_i \in C^1 \quad \text{on } \{x_n = 0\},$$

with A uniformly elliptic and $\|g\|_{L^\infty}$ bounded by the data. The boundary version of De Giorgi-Nash-Moser Theorem gives

$$\|u_i\|_{C^\alpha(B^+_{1/2})} \leq C(\|u_i\|_{L^\infty} + \|g\|_{L^\infty}) \leq C(data).$$

Step 6: C^α estimate near the boundary for the normal derivative

Finally we need to show that u_n satisfies an estimate as above. For this we take a point claim that in a ball of radius $r = y_n/2$ centered at a point $y \in B^+_{1/2}$ we have

$$\fint_{B_r(y)} |\nabla u_n|^2 dx \leq C\, r^{2\alpha-2}. \tag{11.3}$$

Then by the De Giorgi -Nash - Moser theorem in $B_r(y)$ we find

$$[u_n]_{C^\alpha(B_{r/2}(y))} \leq C,$$

and the desired C^α estimate for u_n follows.

To prove (11.3) we notice that from equation (11.2), u_{nn} can be expressed in terms of u_{ij} with $i \neq n$ and a bounded term. Now the claim follows from the Caccioppoli inequality and the Holder continuity of the u_i:

$$\begin{aligned}\fint_{B_r(y)} |\nabla u_i|^2 dx &\leq C\|g\|_{L^\infty} + C\, r^{-2} \fint_{B_{3r/2(y)}} |u_i - \fint u_i|^2\, dx \\ &\leq Cr^{2\alpha-2}.\end{aligned}$$

□

Theorem 11.2. *There exists a unique $u \in C^2(B_1) \cap C^0(\overline{B}_1)$ which solves the Dirichlet problem*

$$\begin{cases} div(\nabla F(\nabla u)) = 0 & \text{in } B_1, \\ u = \varphi \in C^{2,\alpha} & \text{on } \partial B_1, \end{cases}$$

where $F \in C^{2,\alpha}(\mathbb{R}^n)$ with $D^2F > 0$. Moreover, $u \in C^{2,\alpha}(\overline{B}_1)$ and

$$\|u\|_{C^{2,\alpha}(\overline{B}_1)} \le C(n, \alpha, F, \|\varphi\|_{C^{2,\alpha}}).$$

If $F \in C^\infty$ then $u \in C^\infty(B_1)$, and if also $\varphi \in C^\infty$ then $u \in C^\infty(\overline{B}_1)$.

Proof. Uniqueness follows by the maximum principle.

For the existence we use the method of continuity: we show that the set of A of $t \in [0,1]$ for which the Dirichlet problem with boundary data $t\varphi$ has a solution $u_t \in C^{2,\alpha}(\overline{B}_1)$ is both open and closed in $[0,1]$. This set A is nonempty since $u \equiv 0$ solves the equation with $t = 0$. First we assume that $F \in C^4$, and then we remove this restriction.

The fact that A is closed follows by the apriori estimates Theorem 11.1. Indeed, if $t_n \to t_*$, $t_n \in A$, then $\|u_{t_n}\|_{C^{2,\alpha}(\overline{B}_1)} \le C(F, \varphi)$, thus we can extract a convergent subsequence in $C^2(\overline{B}_1)$ to a limit function $u_* \in C^{2,\alpha}(\overline{B}_1)$. Clearly u_* solves the equation with boundary data $t_*\varphi$, hence $t_* \in A$.

The fact that A is open is a consequence of the implicit function theorem on Banach spaces. Indeed

$$T : C^{2,\alpha}(\overline{B}_1) \to C^\alpha(\overline{B}_1) \times C^{2,\alpha}(\partial B_1), \qquad Tu := (div\,(\nabla F(\nabla u)),\ u|_{\partial B_1}),$$

satisfies

$$DTu\ (v) := (div(D^2F(\nabla u)\nabla v), v|_{\partial B_1}),$$

which is an isomorphism between the domain and the target by the Schauder estimates for linear equations.

Finally, we can remove the restriction that $F \in C^4$ by approximating F by $F_m \in C^4$ and by solving the corresponding Dirichlet problems for F_m, and then taking the limit.

∎

Remark: 1) The approximation result shows that if $\varphi, F \in C^{1,1}$ and D^2F is uniformlly elliptic on compact sets of $\mathbb{R}^n$, then we can solve the Dirichlet problem in $C^{1,\alpha}(\overline{B}_1)$. The solution u is understood in the weak sense and it minimizes the energy $\int F(\nabla v)dx$ among all Lipschitz functions with the same boundary values.

2) The Dirichlet problem in more general (nonconvex) domains and with nonzero right had sides might not be solvable, see Problems 1,2 below. The main obstacles are with Steps 1 and 2, (Steps 3-6 can be easily extended) .

11.1. **Problems.**

1. Show that the Dirichlet problem for the minimal surface equation

$$div(\nabla F(\nabla u)) = 0 \quad \text{in } B_2 \setminus B_1,$$

$$u = 0 \quad \text{on } \partial B_2, \quad u = C \quad \text{on } \partial B_1,$$

is not solvable if C is large. Find a condition on the domain Ω so that Step 2 in the proof of Theorem 11.1 can be implemented.

2. Show that the minimal surface equation

$$div(\nabla F(\nabla u)) = c \quad \text{in } B_1, \qquad u = 0 \quad \text{on } \partial B_1,$$

is not solvable if $|c|$ is large. Find the optimal values of c for solvability.

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